

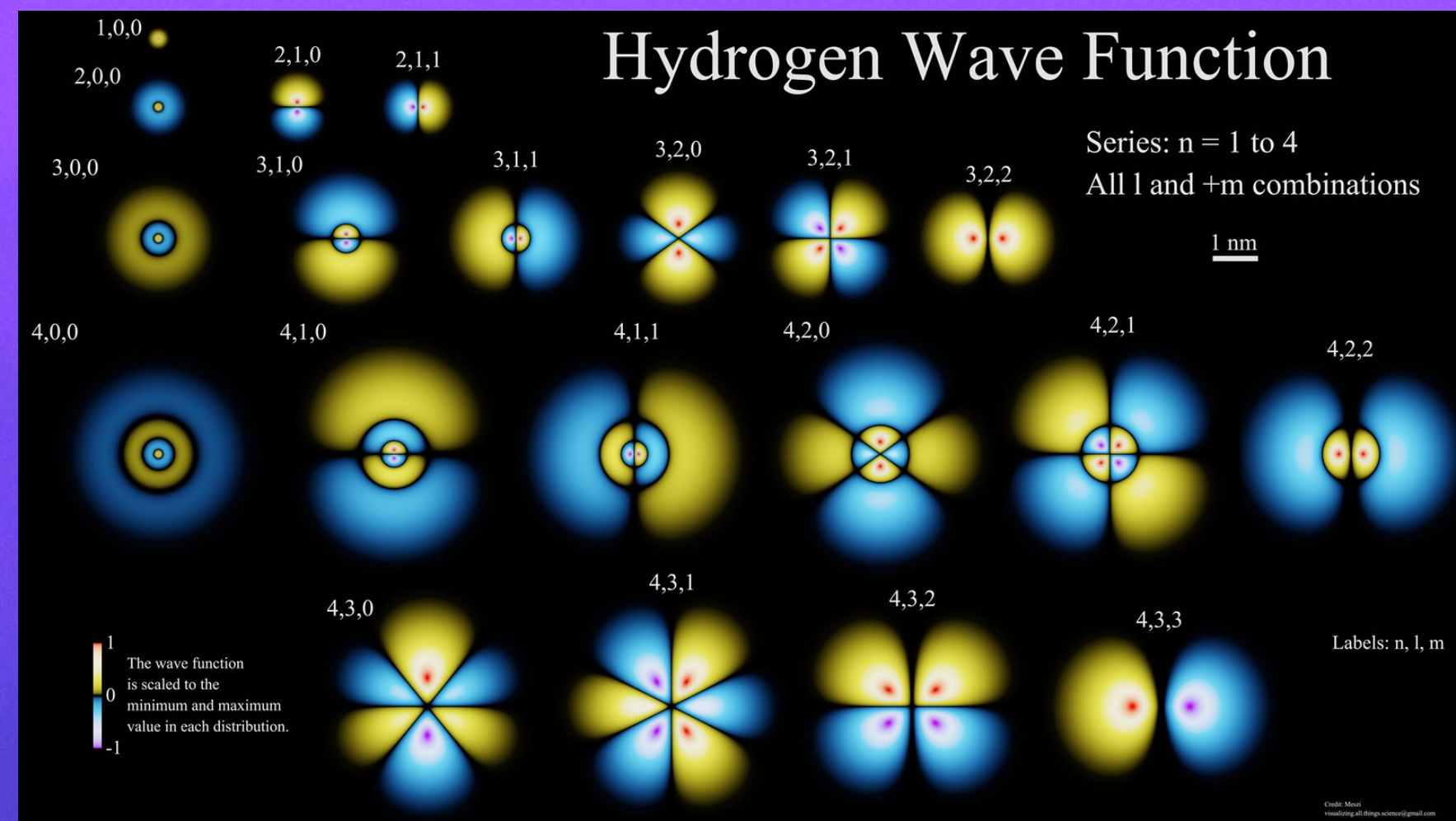
DFT with periodic bordering conditions

Bands Structure and Convergences.

Theoretical Fundaments

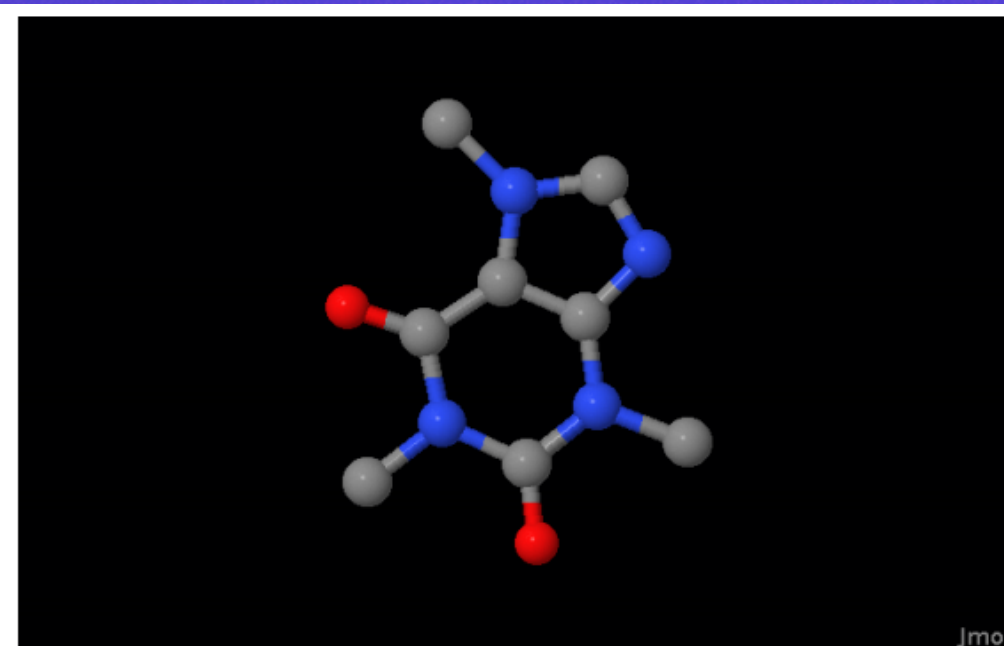
Density Functional Theory

The Perfect Balance: Cost vs. Accuracy

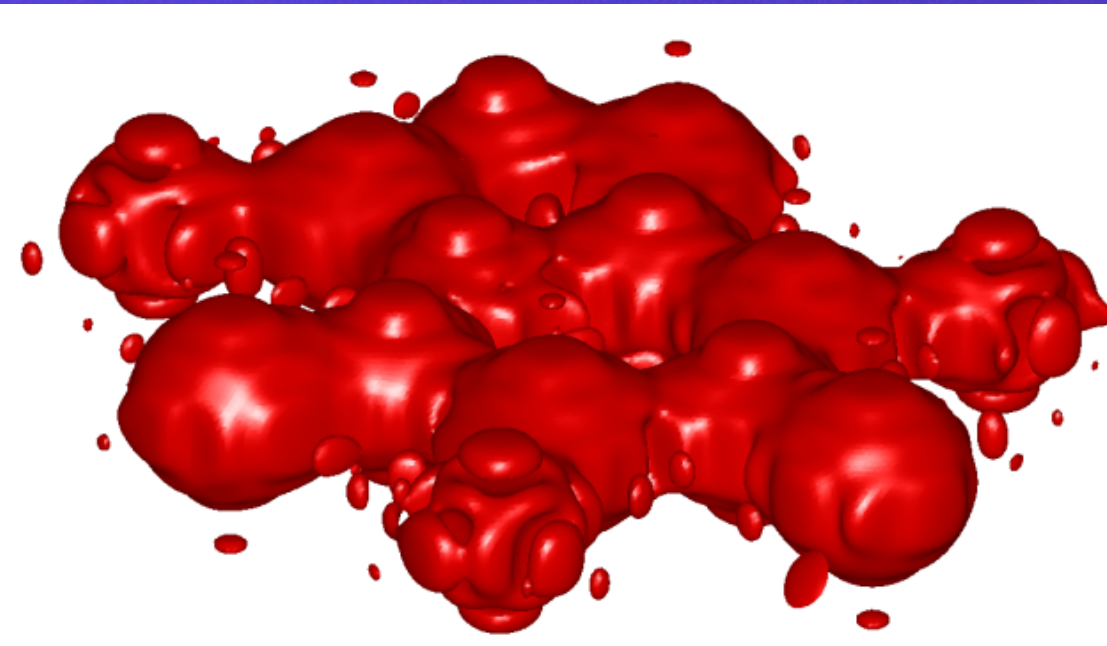


The Goal is Density: Instead of searching for the N -electron wave function, DFT proves that the electron density contains all the system's information.

The Kohn-Sham "Shortcut": A fictitious system of non-interacting electrons is solved, which by design has the same density as the real system.



(a) Schematic representation

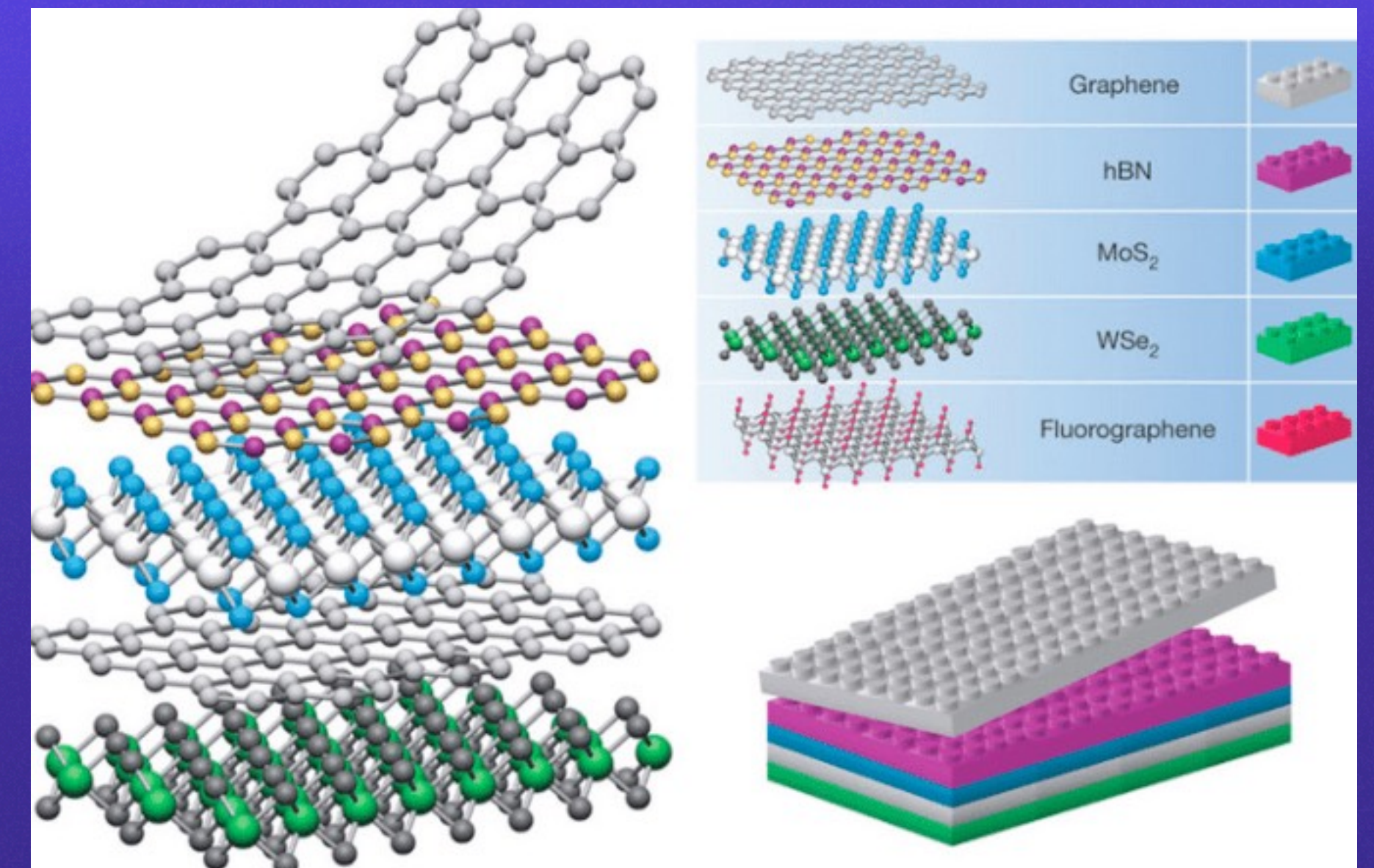
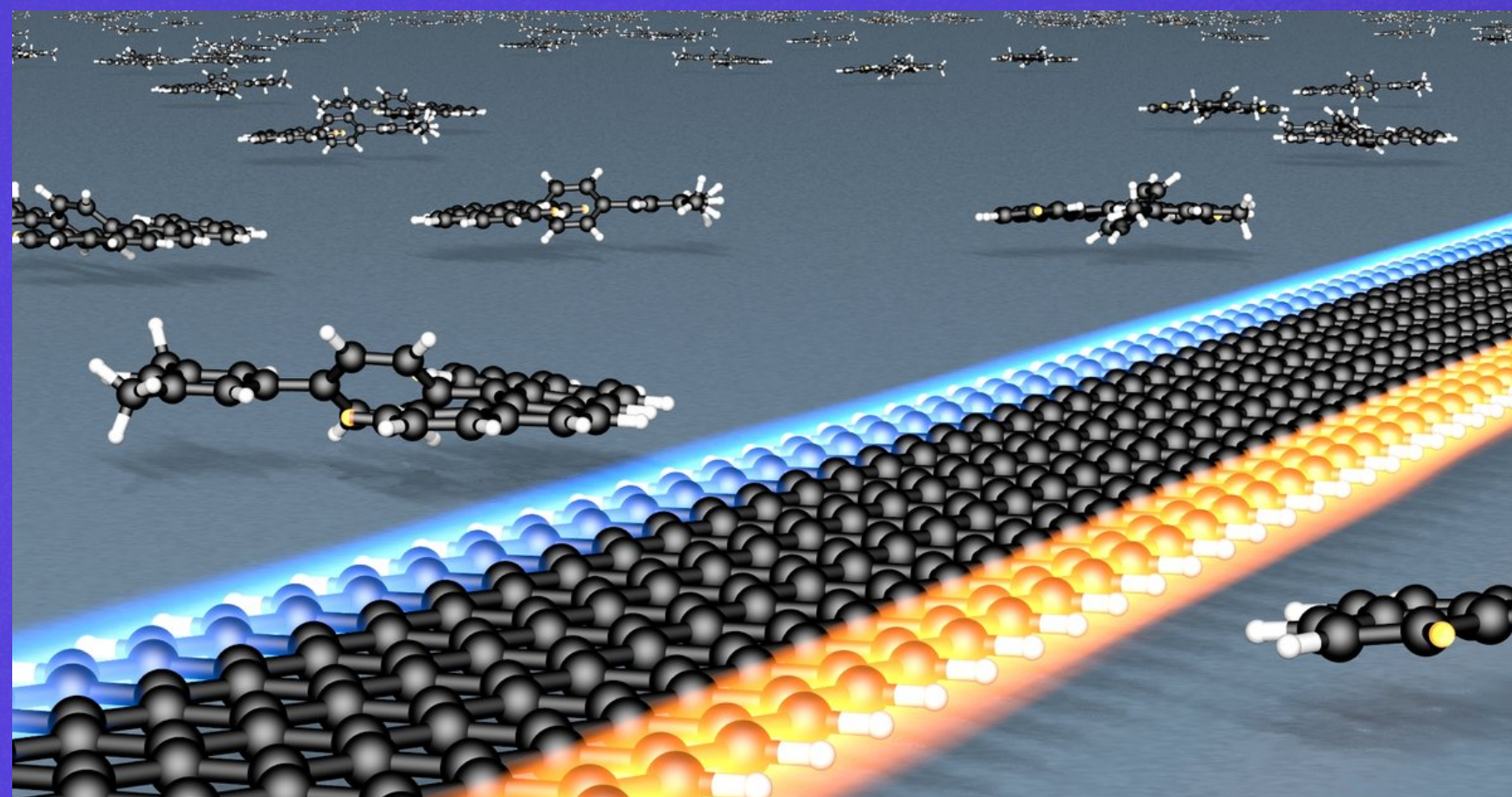
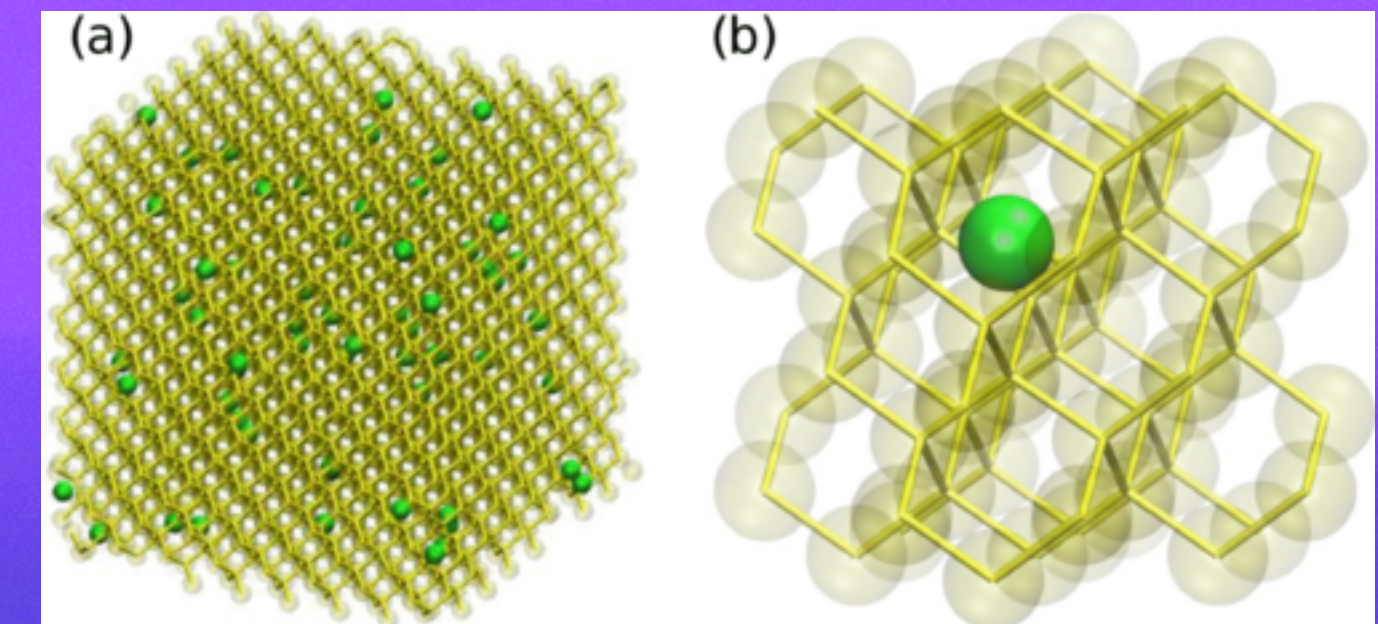
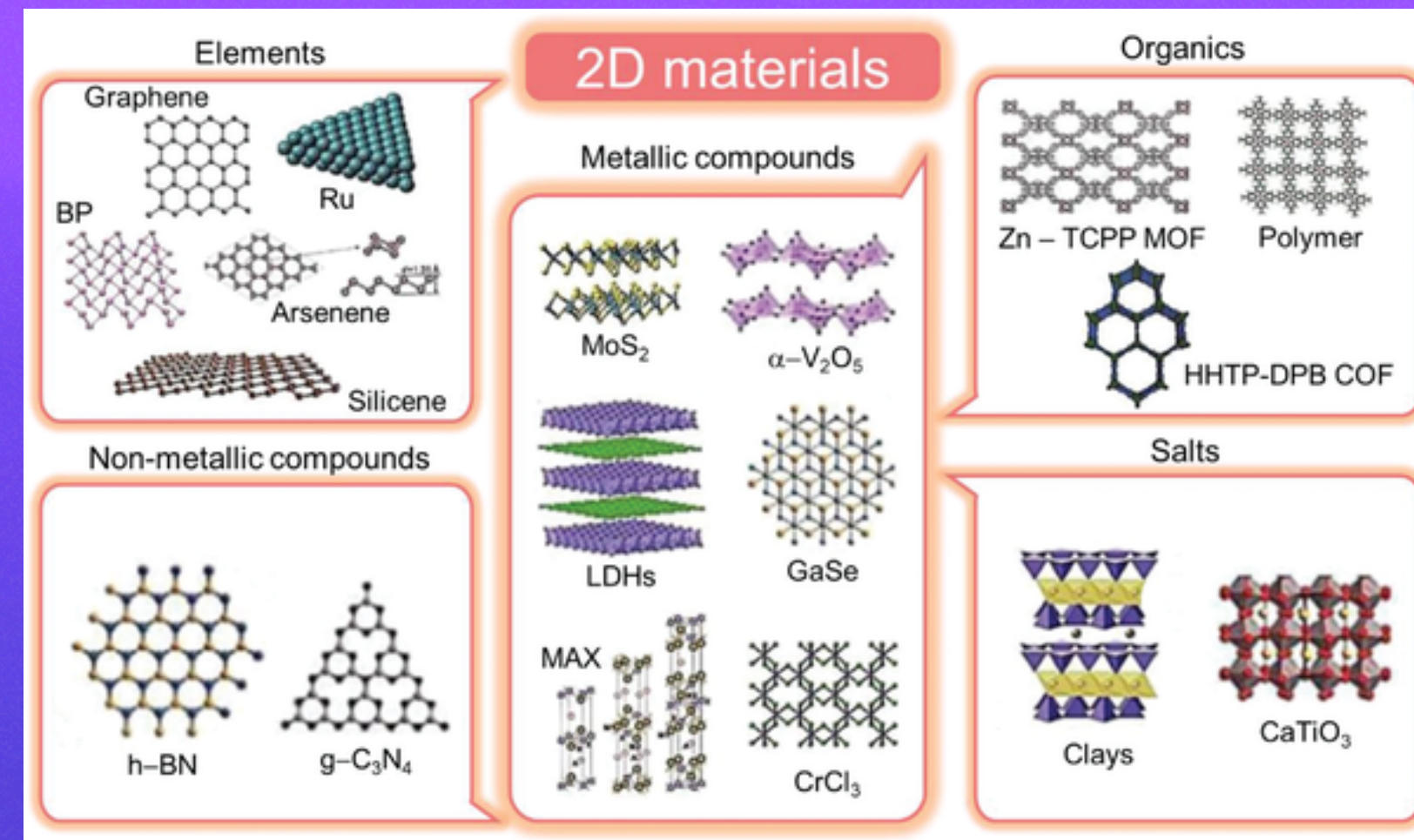
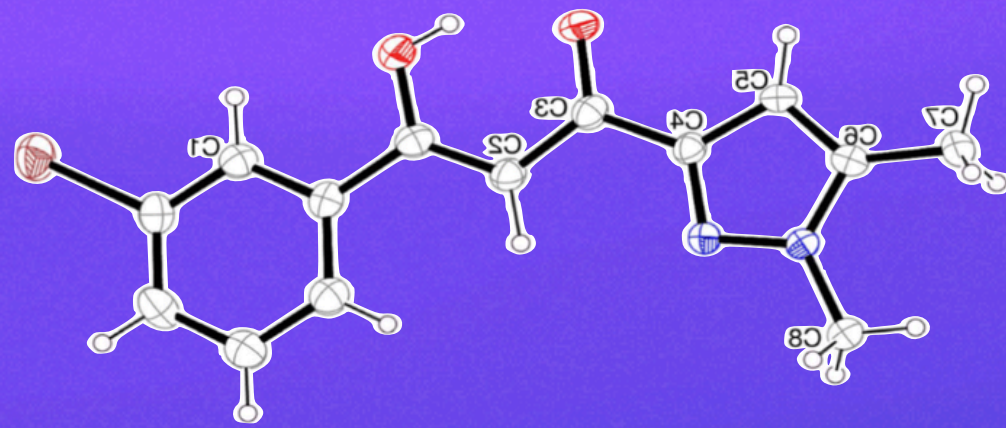


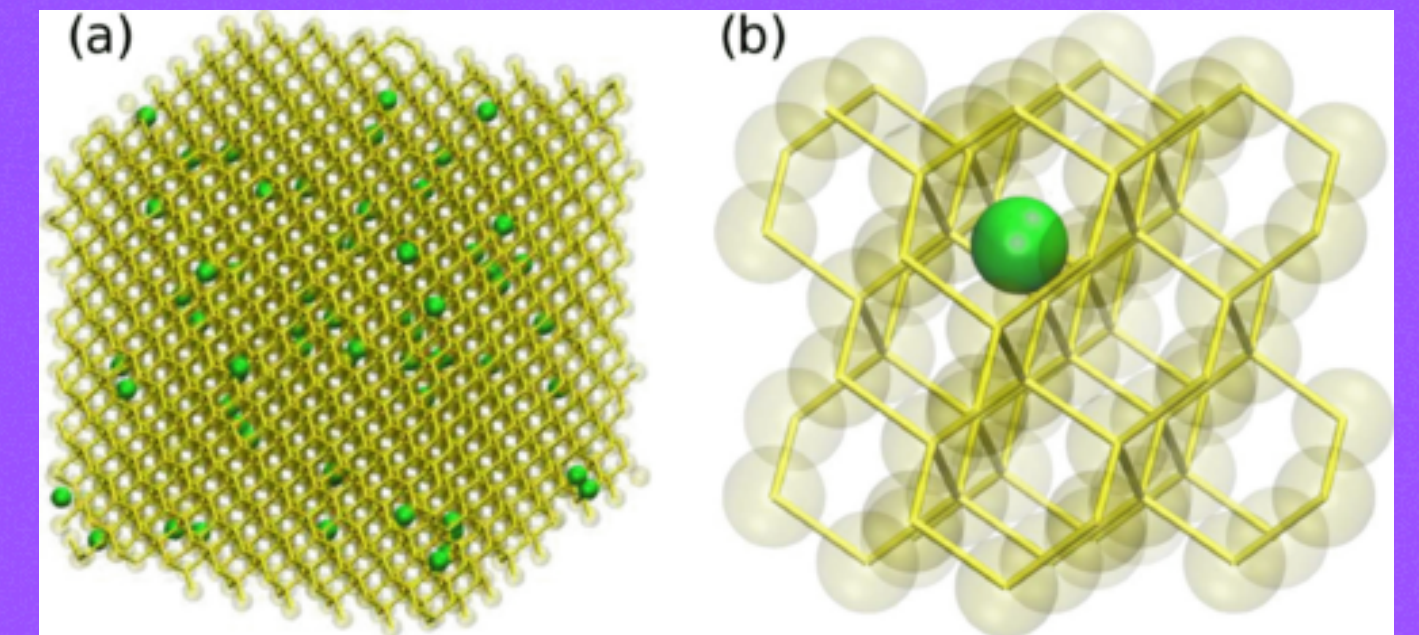
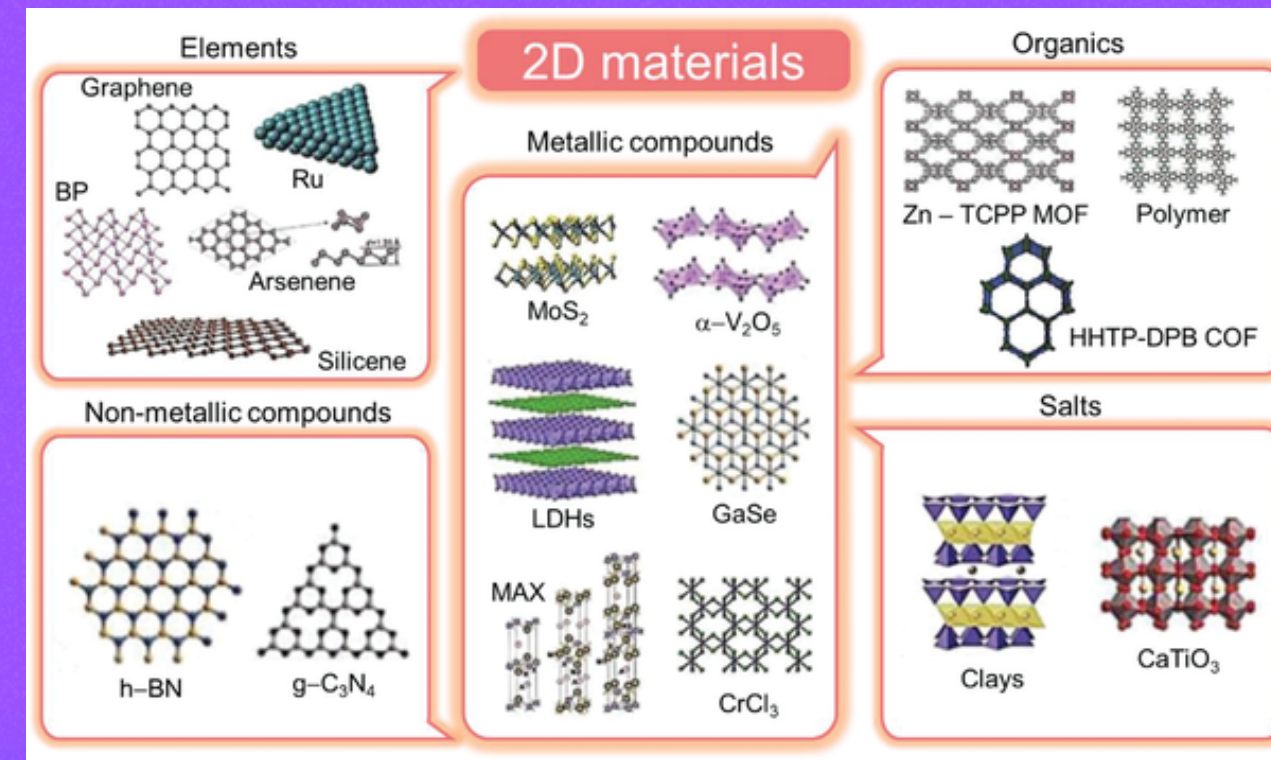
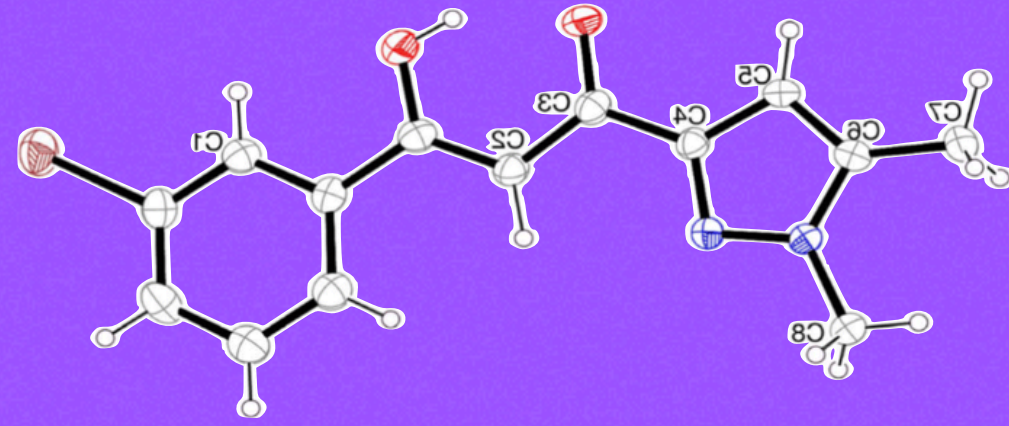
(b) Electron density map

All the complex quantum physics (exchange and correlation) that Post-HF methods calculate explicitly and expensively, DFT groups into a single term: **the exchange-correlation functional.**

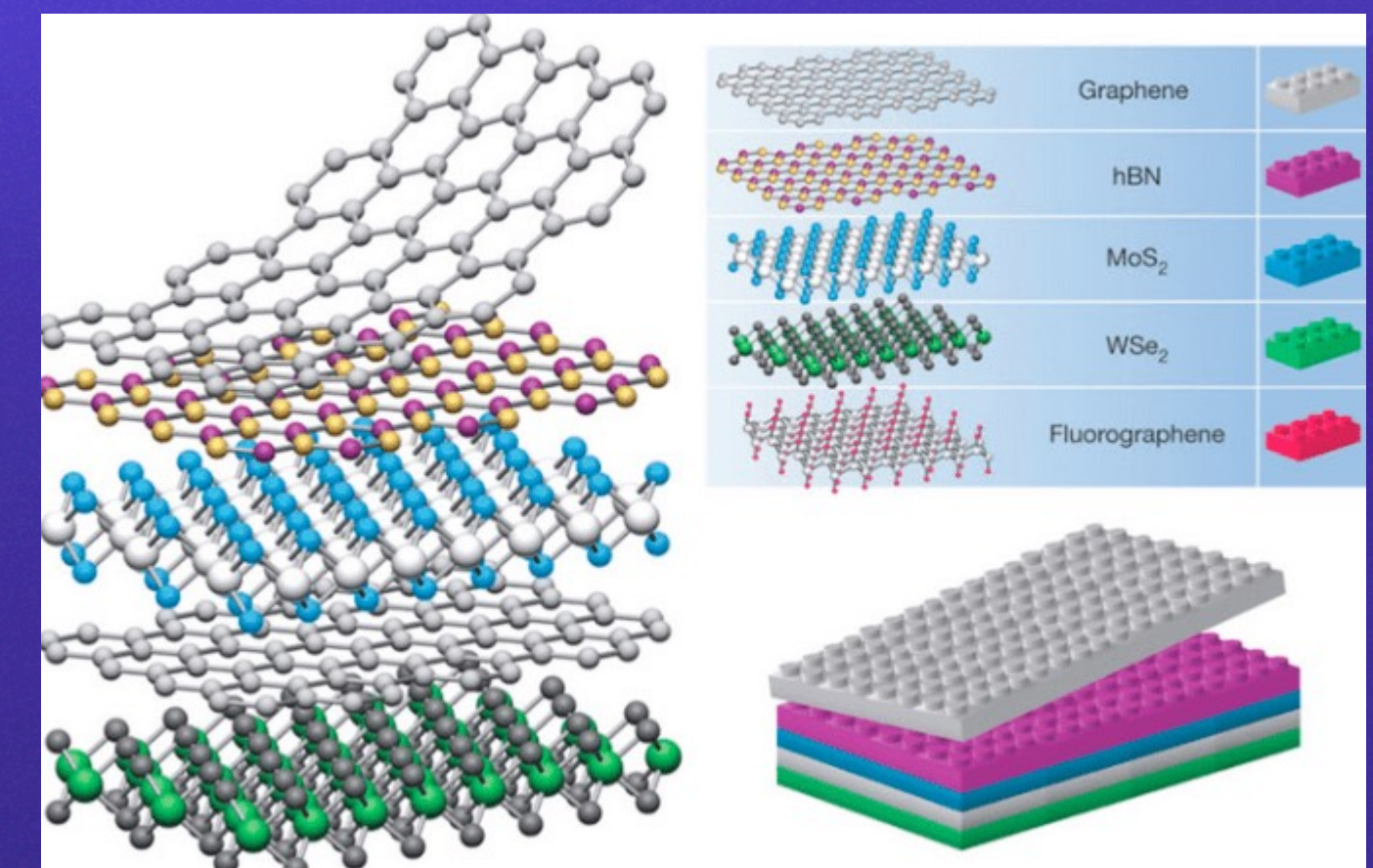
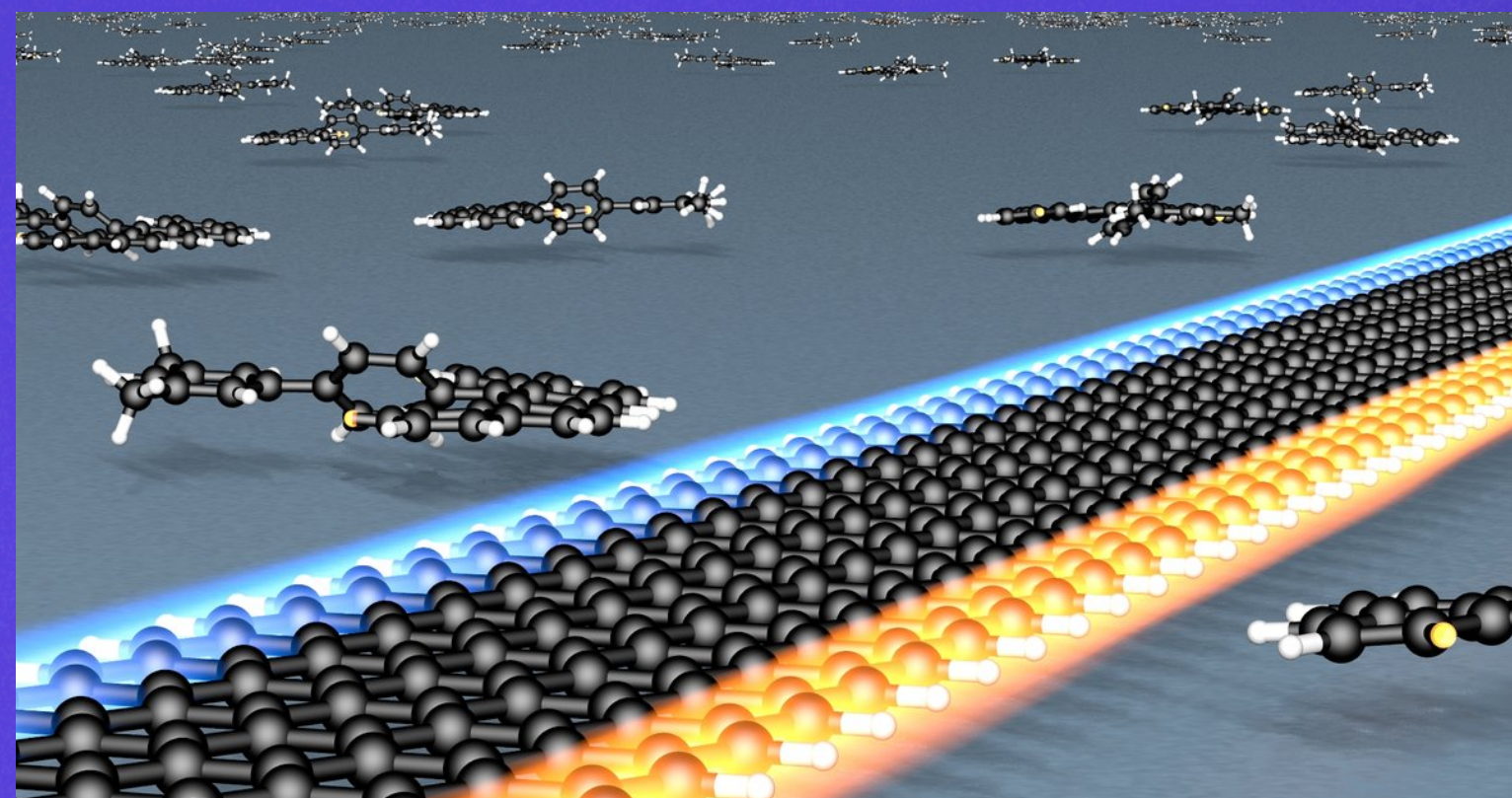
DFT & Periodic Materials

From molecules to crystal lattices





What is different in a solid state?



DFT & Periodic Materials

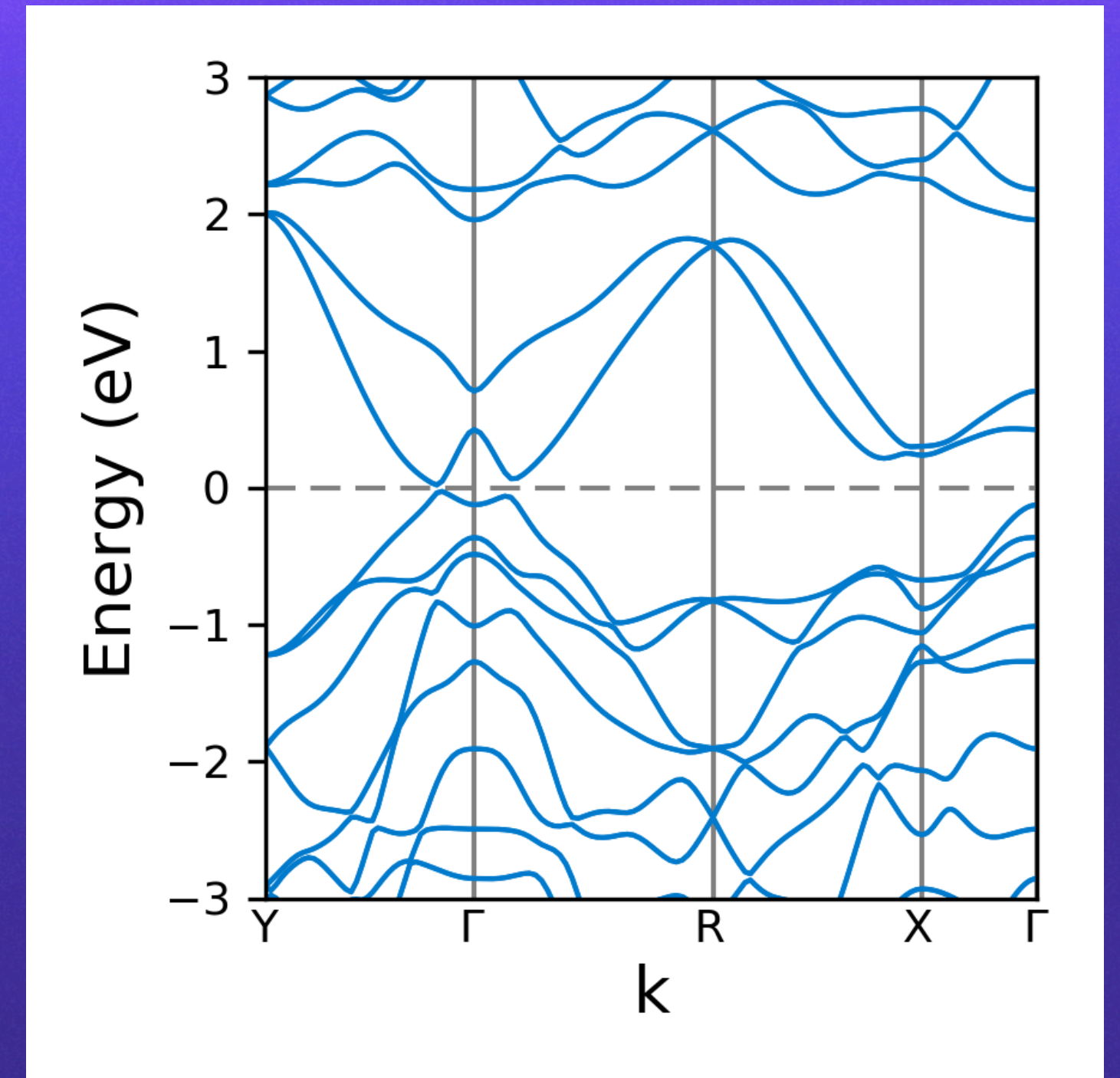
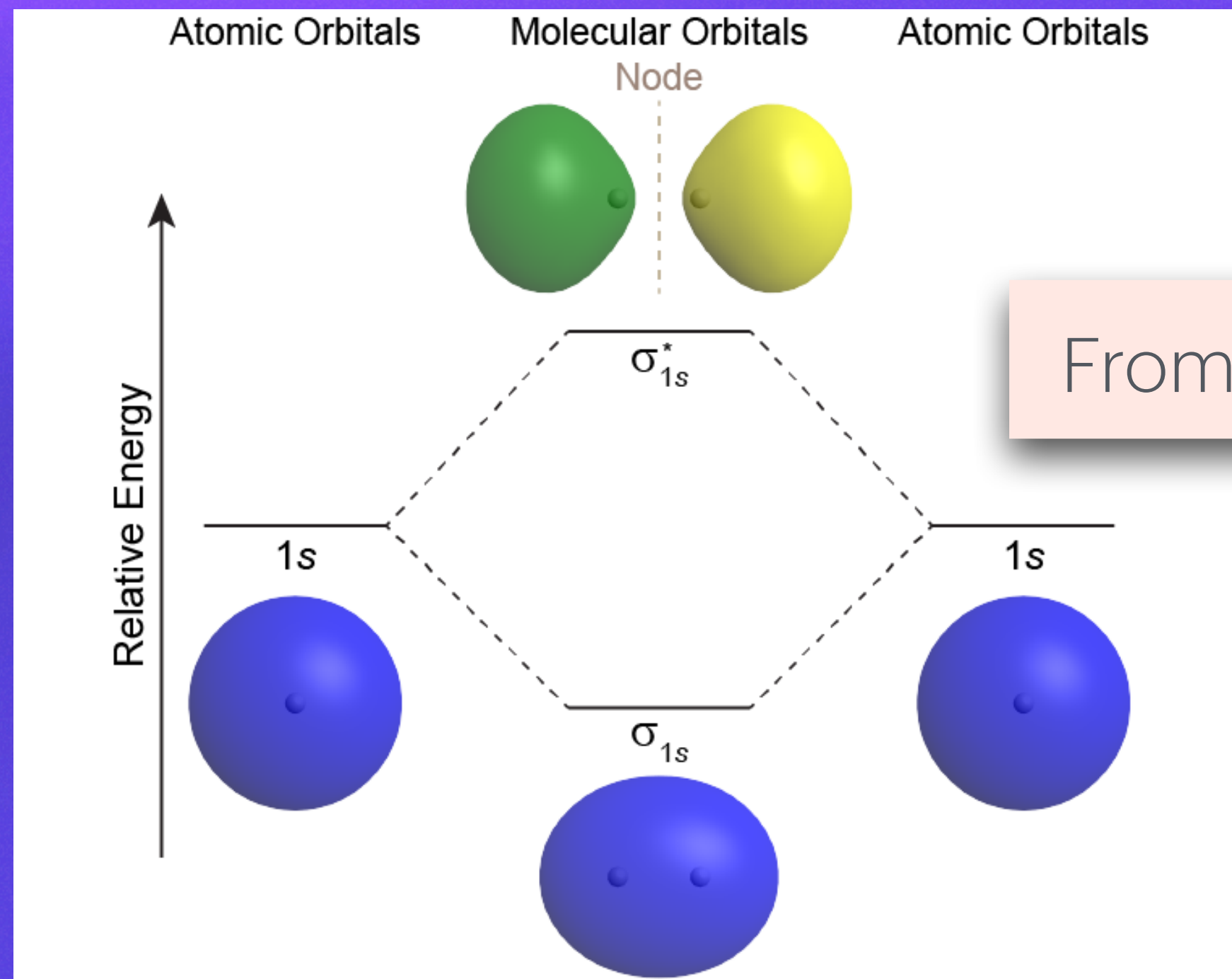
From molecules to crystal lattices

Pseudopotentials

Reciprocal Space

Bloch Theorem

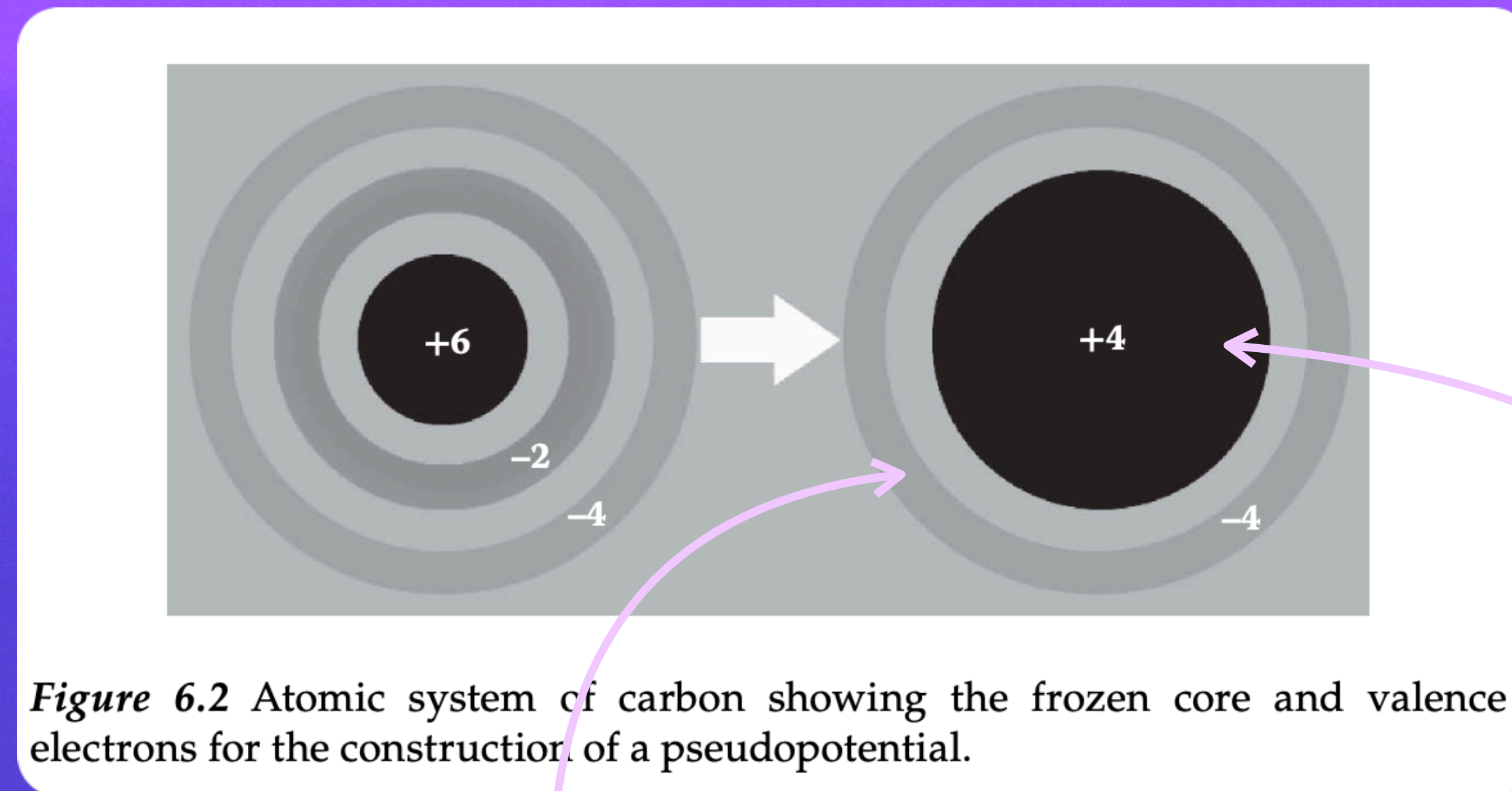
From orbitals to band structure



Pseudopotentials

DFT & Periodic Materials

Pseudopotentials (PPs)



Forming bonds, being ionized,
conducting electricity in metals, and
performing other atomic activities

Freezing the nucleus and
the core electrons
together

Intro DFT & Materiales periódicos

Pseudopotentials (PPs)

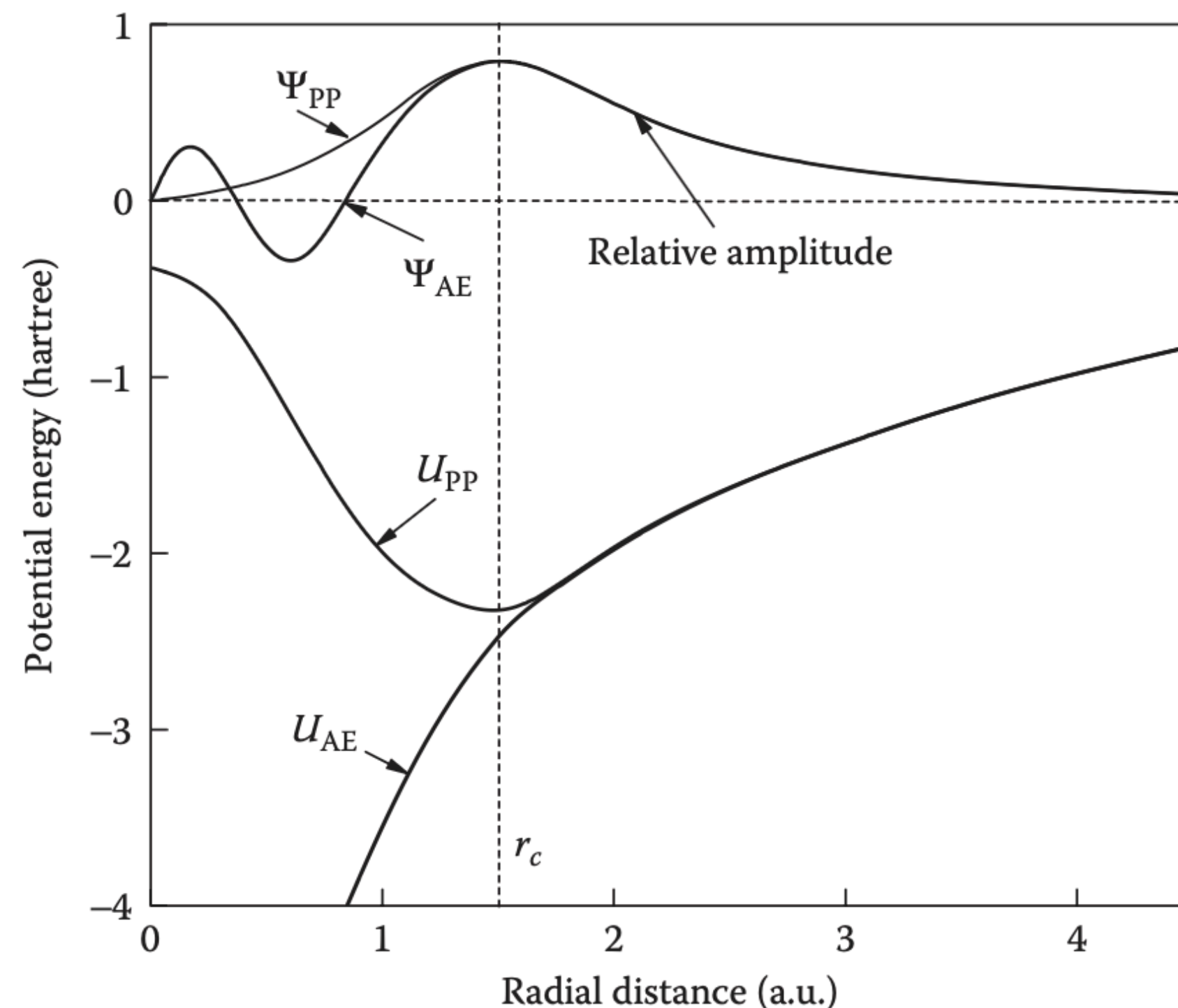


Figure 6.3 Schematic illustration of a pseudo wave function pseudized from a 3s wave function (showing the relative amplitude in arbitrary unit) and the corresponding pseudo- and all-electron (AE) potentials.

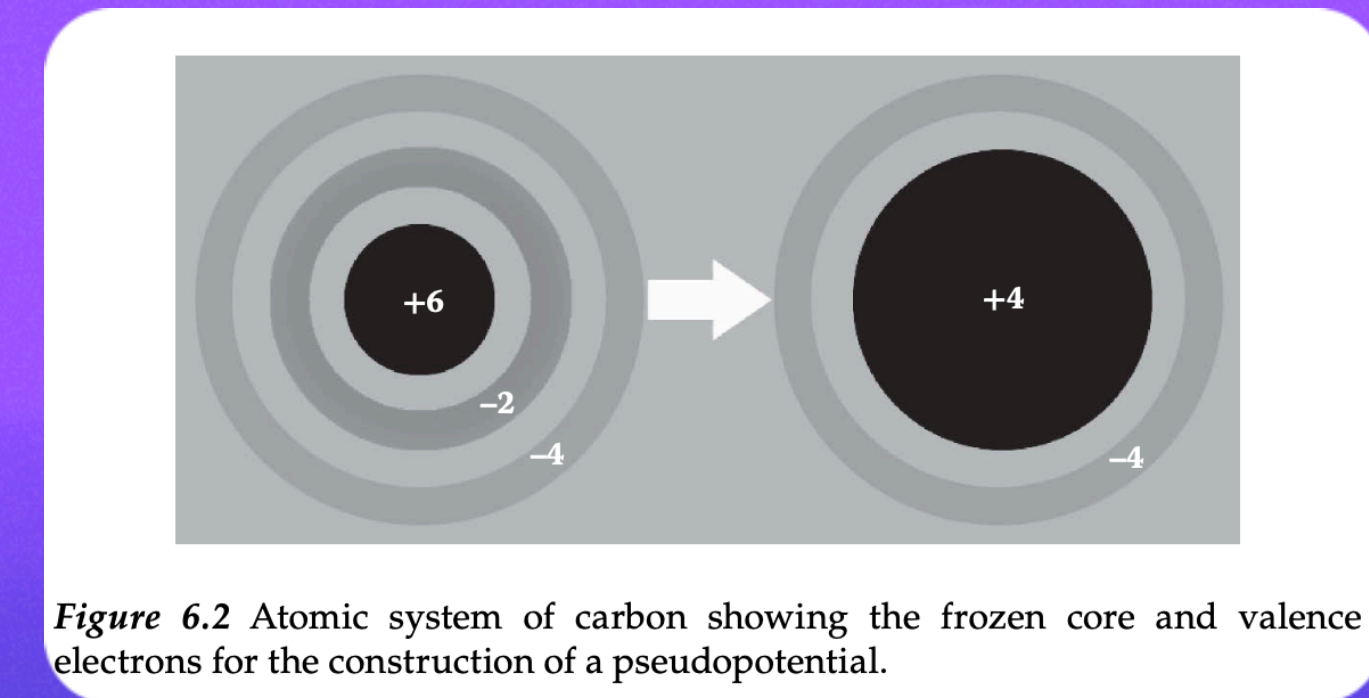


Figure 6.2 Atomic system of carbon showing the frozen core and valence electrons for the construction of a pseudopotential.

- Markedly reduces the number of PWs needed for the expansion of the pseudo-wavefunction.
- Allows for more noticeable energy changes.
- Eliminates the relativistic effect of core electrons.

Intro DFT & Materiales periódicos

Pseudopotentials (PPs)

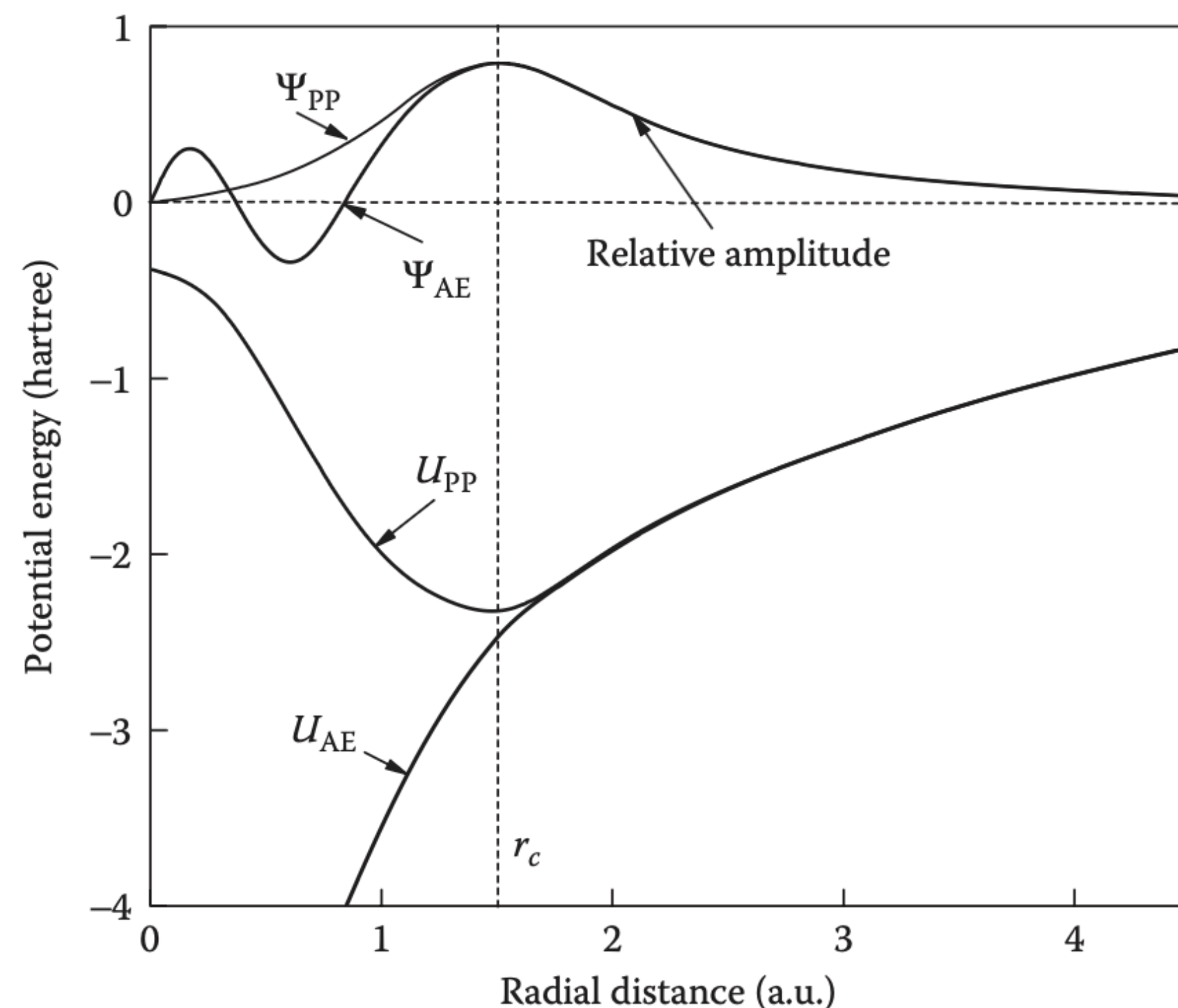


Figure 6.3 Schematic illustration of a pseudo wave function pseudized from a 3s wave function (showing the relative amplitude in arbitrary unit) and the corresponding pseudo- and all-electron (AE) potentials.

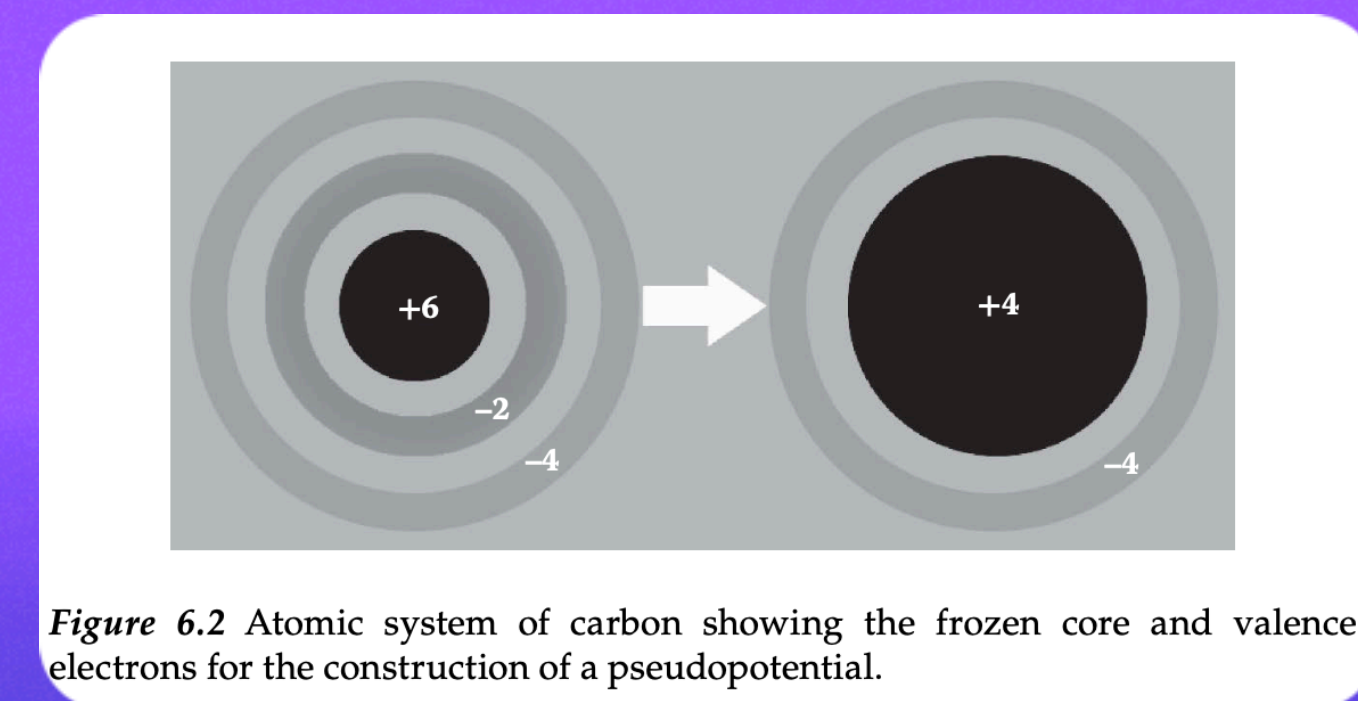


Figure 6.2 Atomic system of carbon showing the frozen core and valence electrons for the construction of a pseudopotential.

$$\left[-\frac{1}{2} \nabla^2 + U_{PP}[\rho(\mathbf{r})] \right] \psi_i^{PP}(\mathbf{r}) = \epsilon_i \psi_i^{PP}(\mathbf{r})$$

$$\rho(\mathbf{r}) = \sum_i |\psi_i^{PP}(\mathbf{r})|^2$$

Intro DFT & Materiales periódicos

Pseudopotentials (PPs)

Norm-Conserving (NCPPs):

Ultrasoft (USPPs):

Projector Augmented-Wave (PAW):

Intro DFT & Materiales periódicos

Pseudopotentials (PPs)

Norm-Conserving (NCPPs):

Ultrasoft (USPPs):

Projector Augmented-Wave (PAW):

- It ensures that the total integrated charge within a defined radius (the "core radius") is identical to that of the all-electron (AE) atom.
- It guarantees that, outside of this radius, the valence electrons feel the exact same potential.

Intro DFT & Materiales periódicos

Pseudopotentials (PPs)

Norm-Conserving (NCPPs):

Ultrasoft (USPPs):

Projector Augmented-Wave (PAW):

- Relaxes the norm-conservation rule to create much "softer" (smoother) wavefunctions.
- Drastically reduces the required number of plane waves.

Intro DFT & Materiales periódicos

Pseudopotentials (PPs)

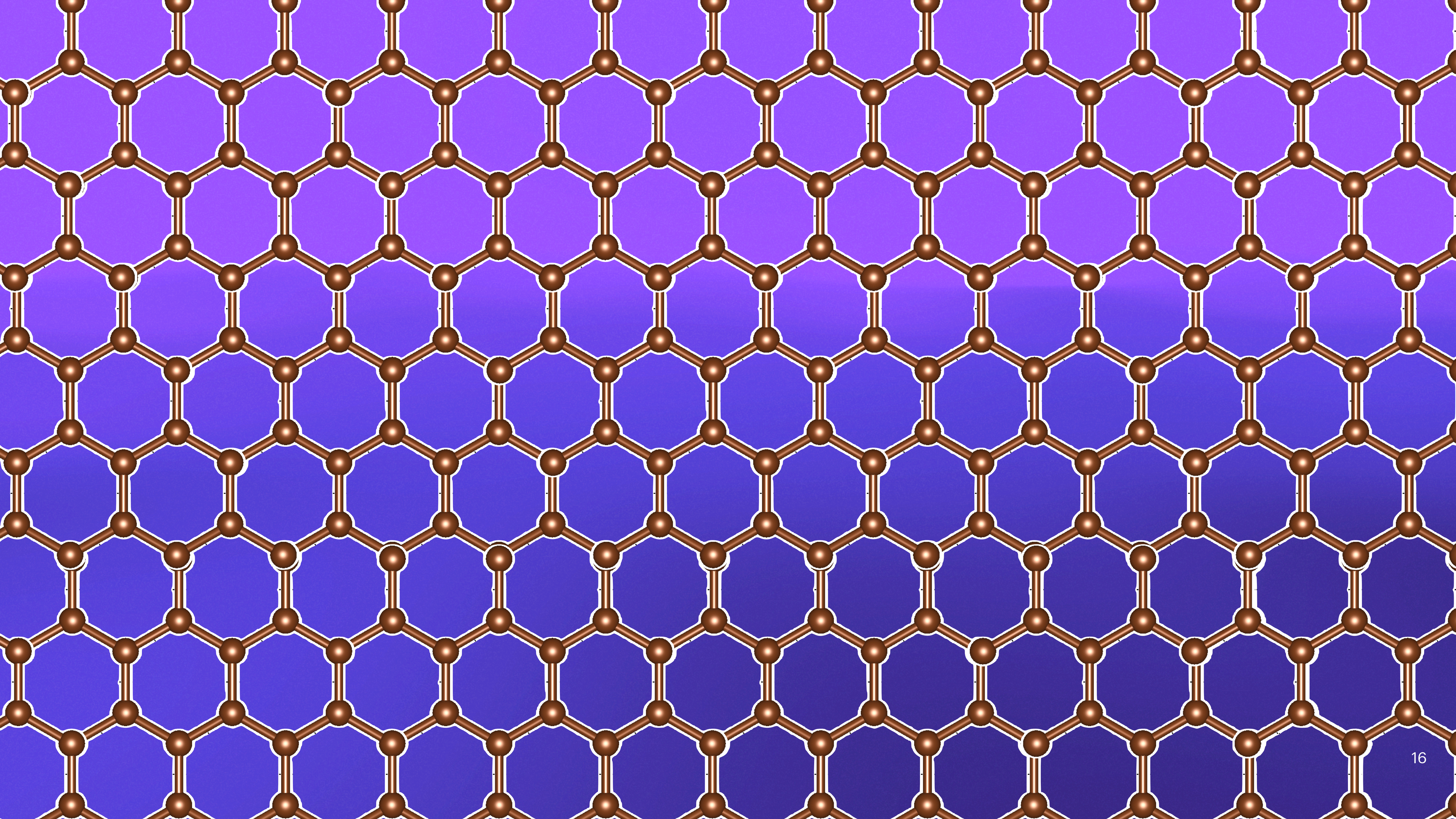
Norm-Conserving (NCPPs):

Ultrasoft (USPPs):

Projector Augmented-Wave (PAW):

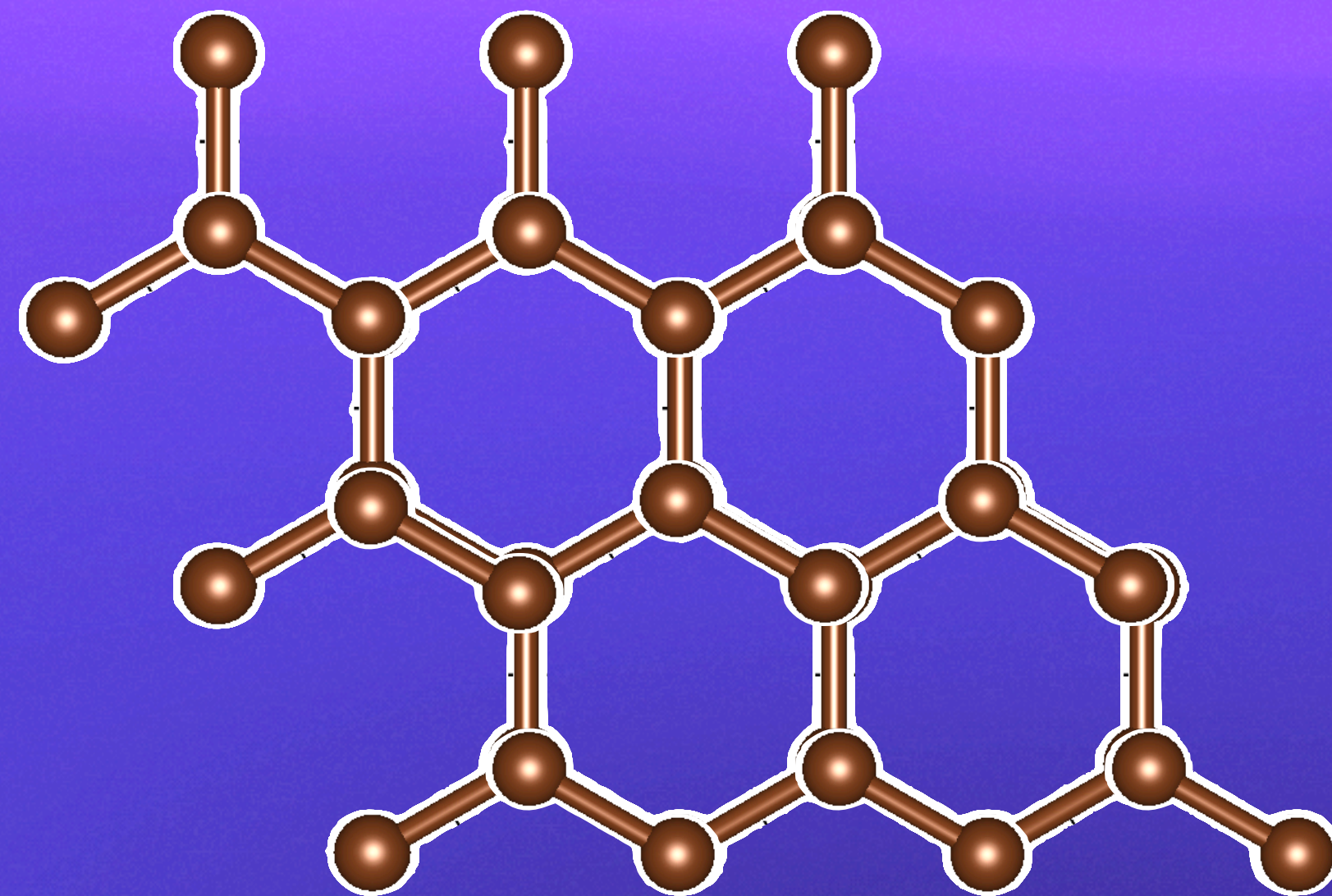
- Combines the efficiency of ultrasoft potentials with the accuracy of an all-electron calculation.
- Uses a transformation to reconstruct the exact wavefunctions near the core.

Symmetry and Periodicity



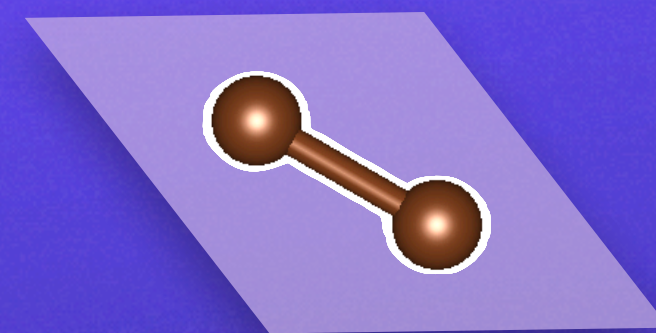
Symmetry and Periodicity

Unit Cell - Supercell



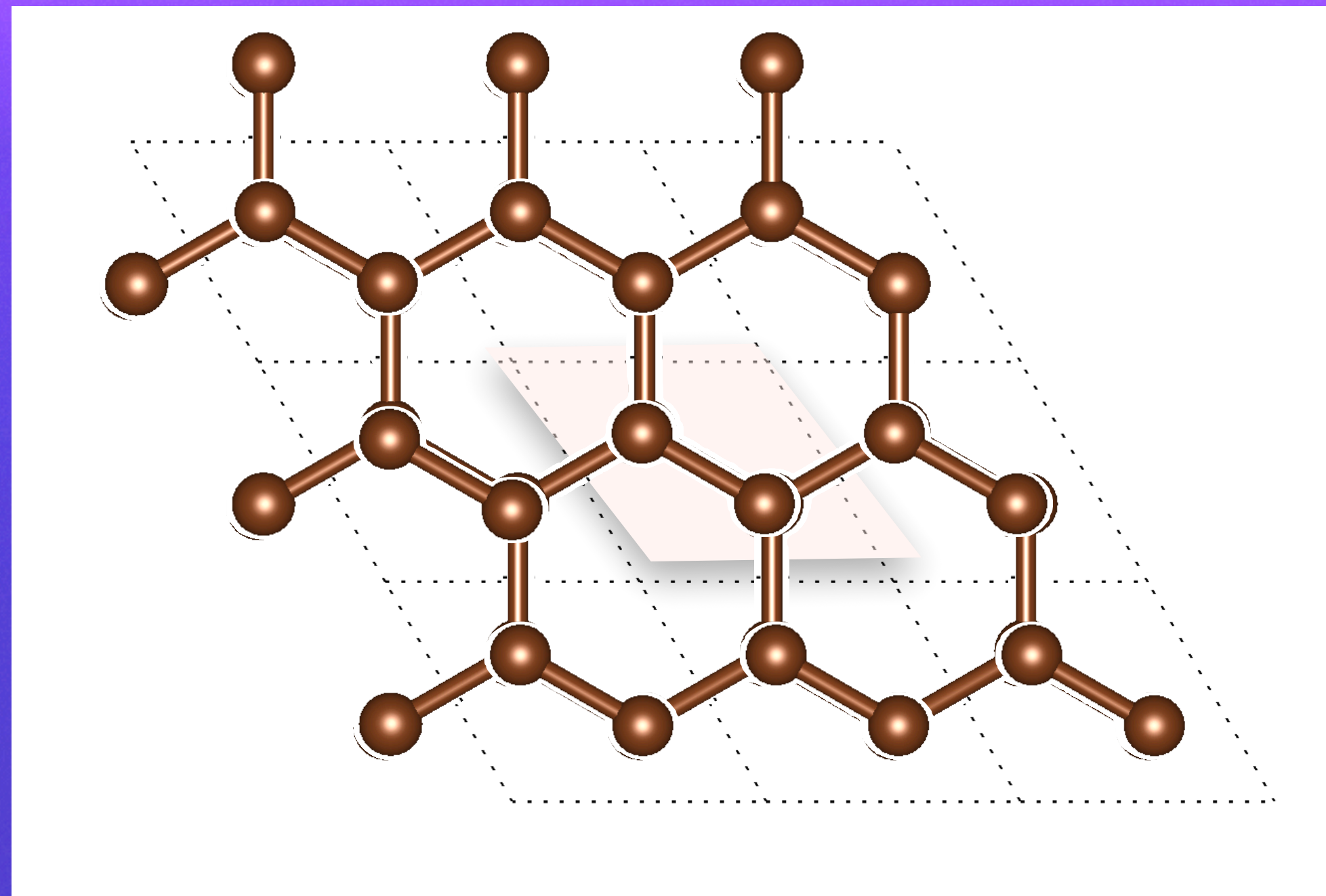
Symmetry and Periodicity

Unit Cell - Supercell



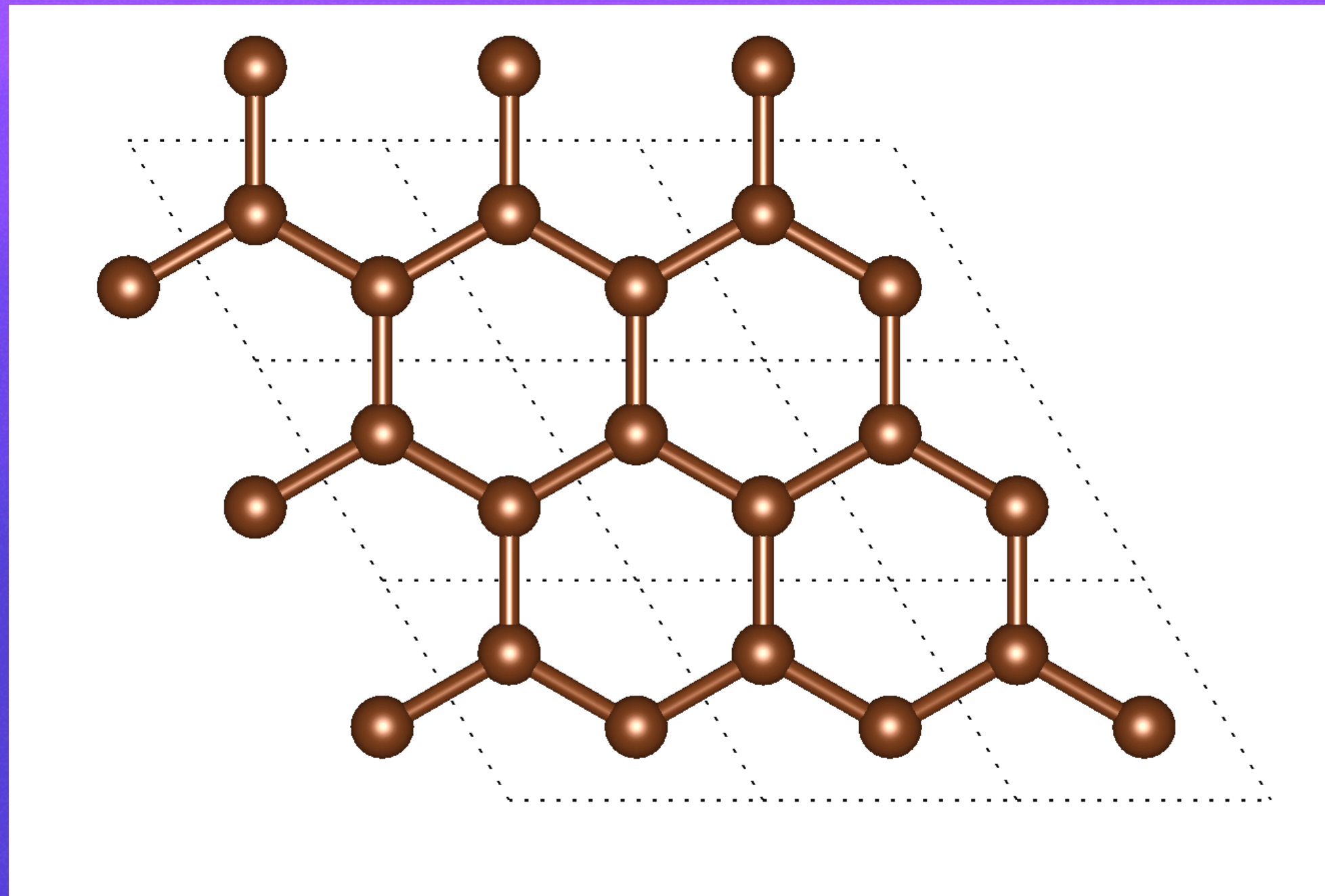
Symmetry and Periodicity

Unit Cell - Supercell

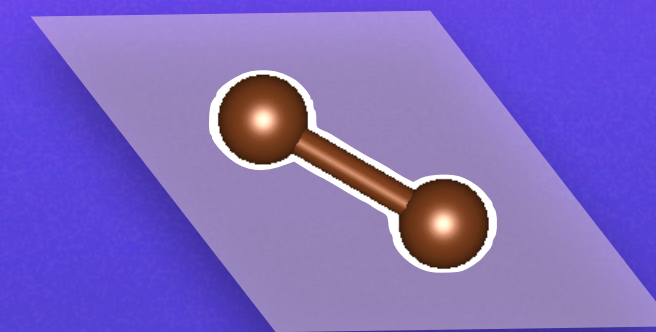


Symmetry and Periodicity

Unit Cell - Supercell



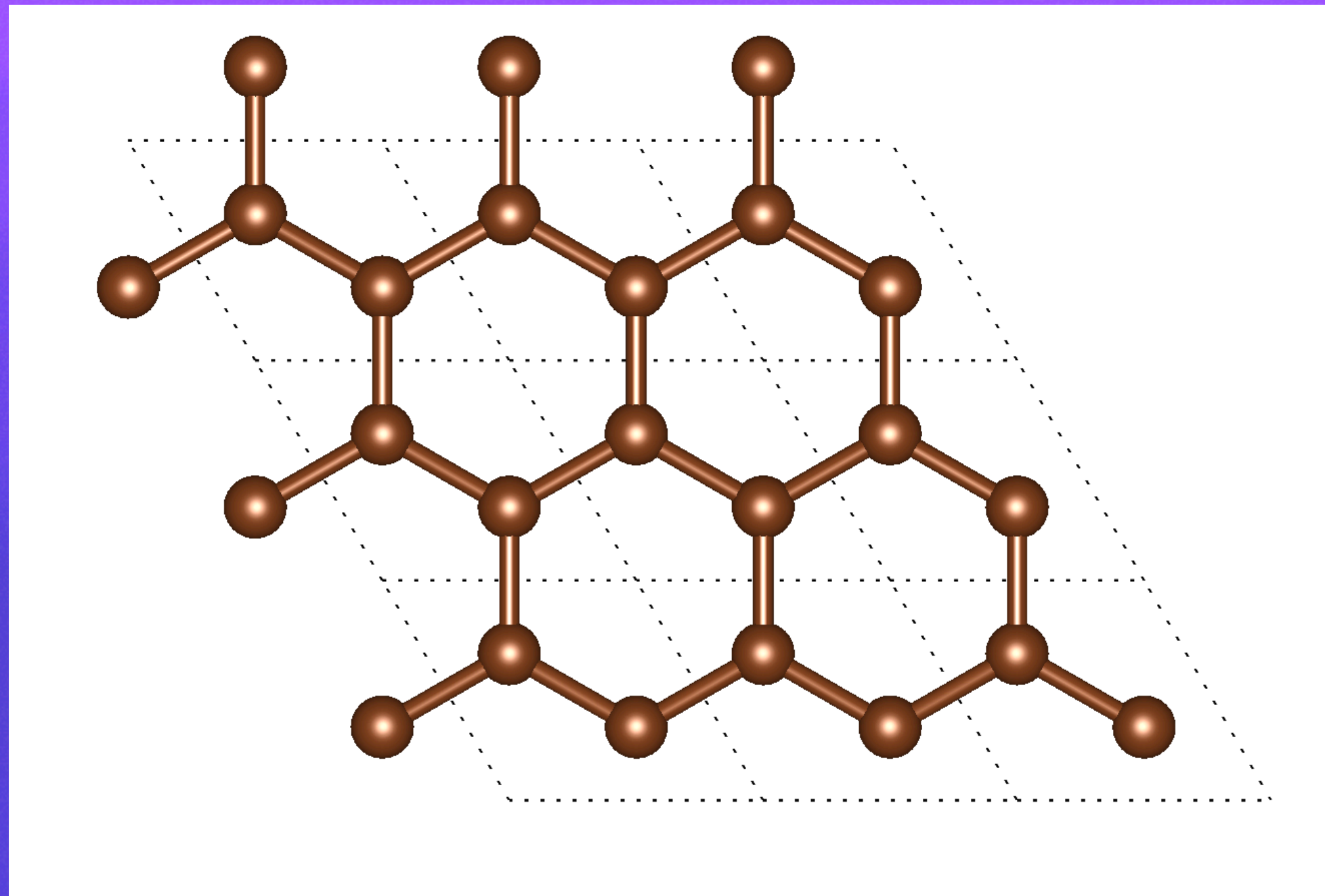
Supercell 3x3



Unit Cell: The minimal representation of the system.

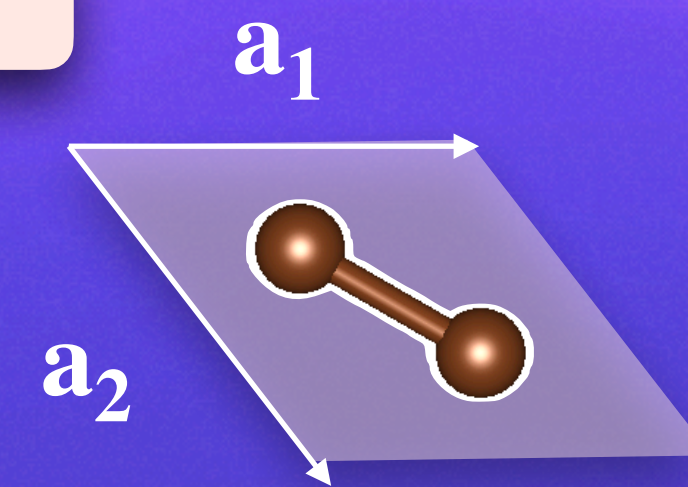
Symmetry and Periodicity

Unit Cell - Supercell



Supercell 3x3

Primitive basis



Unit Cell: The minimal representation of the system.

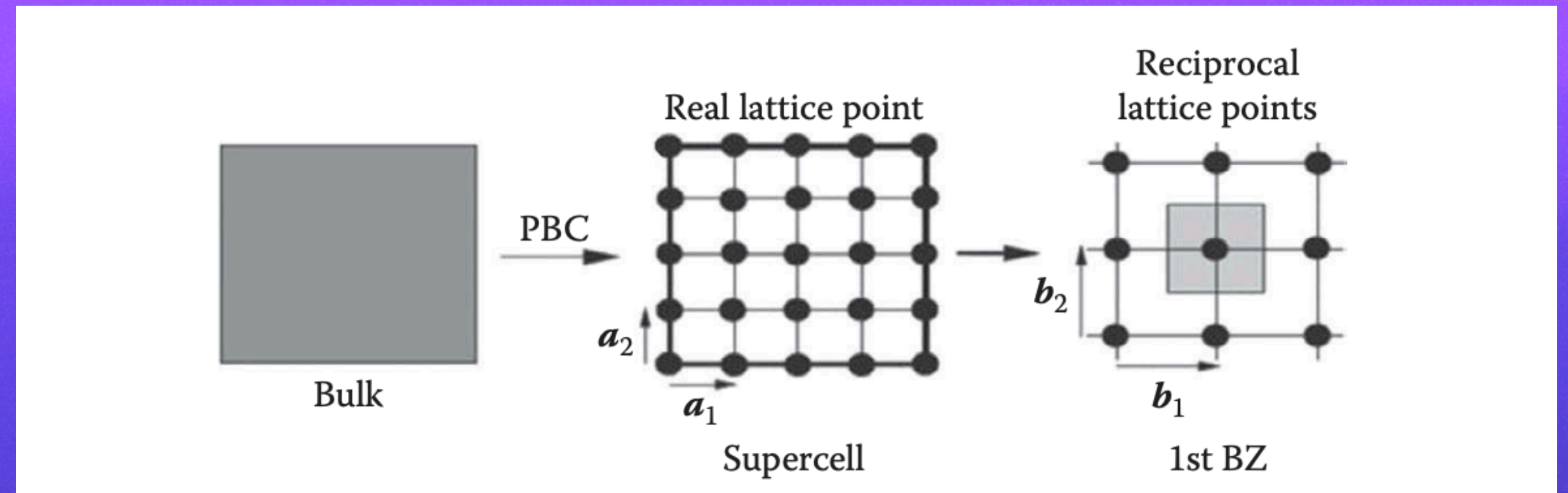
Bravais lattice

$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3$$

Symmetry and Periodicity

Reciprocal Space

Mathematical transformation of the crystal lattice into a new "space" which describes **periodicity and wave vectors $\mathbf{K}$** .



$$e^{i\mathbf{K}\cdot\mathbf{r}} = e^{i\mathbf{K}\cdot(\mathbf{r}+\mathbf{R})}$$

$$\vec{b}_1 = 2\pi \frac{\vec{a}_2 \times \vec{a}_3}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} \quad \vec{b}_2 = 2\pi \frac{\vec{a}_3 \times \vec{a}_1}{\vec{a}_2 \cdot (\vec{a}_3 \times \vec{a}_1)} \quad \vec{b}_3 = 2\pi \frac{\vec{a}_1 \times \vec{a}_2}{\vec{a}_3 \cdot (\vec{a}_1 \times \vec{a}_2)}$$

Reciprocal space vectors

$$\mathbf{K} = k_1 \mathbf{b}_1 + k_2 \mathbf{b}_2 + k_3 \mathbf{b}_3$$

Symmetry and Periodicity

Reciprocal Space- Brillouin Zone

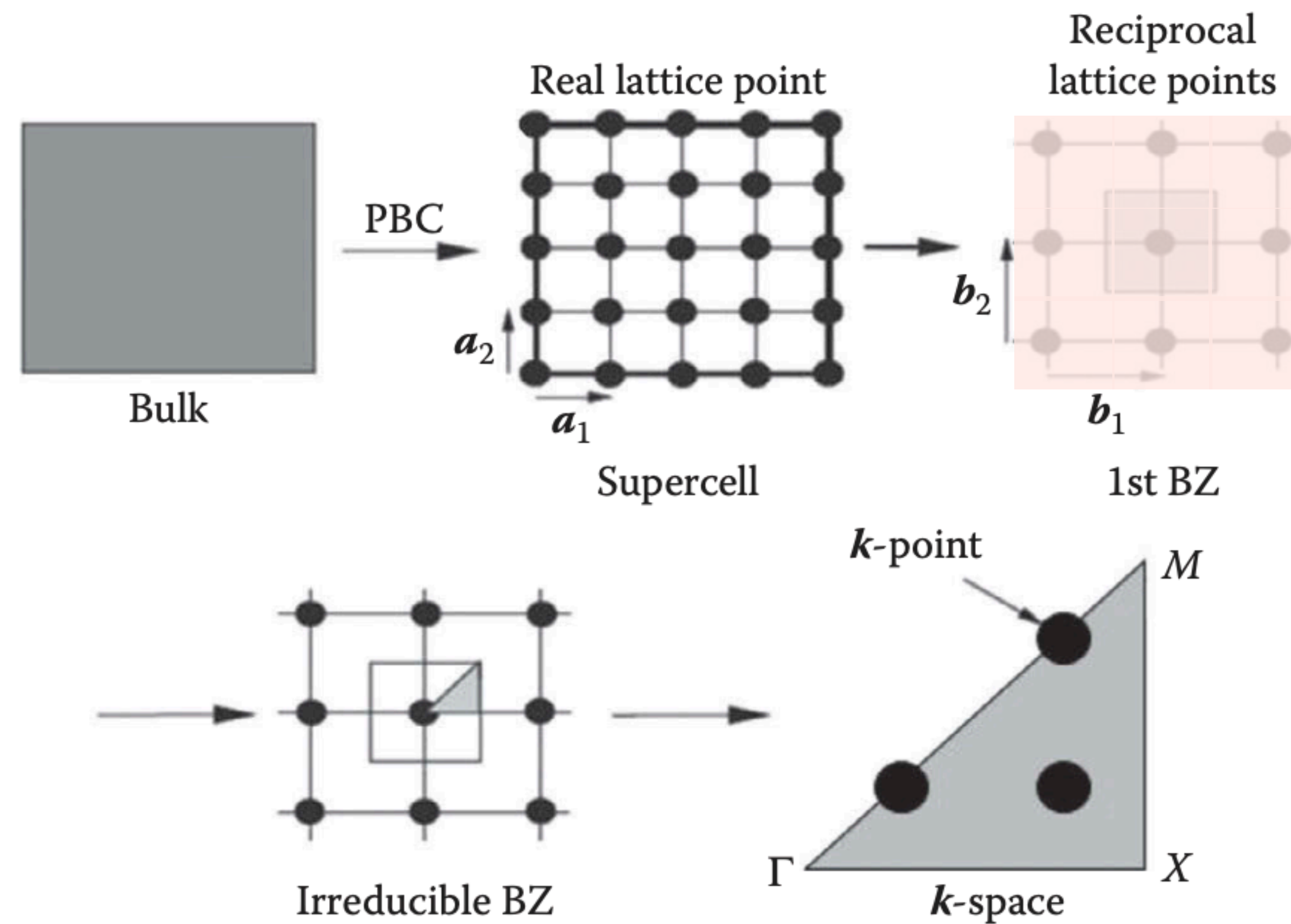
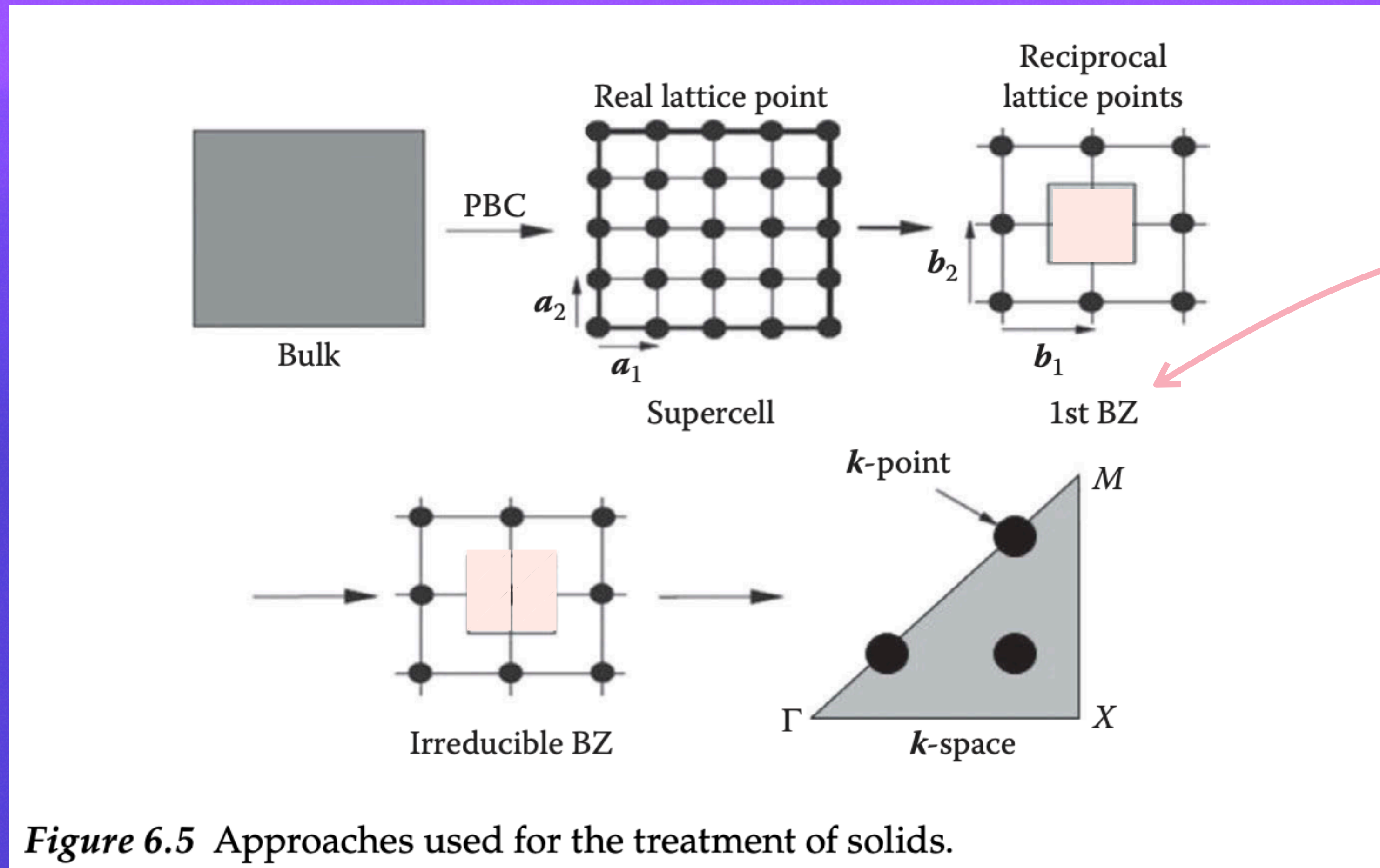


Figure 6.5 Approaches used for the treatment of solids.

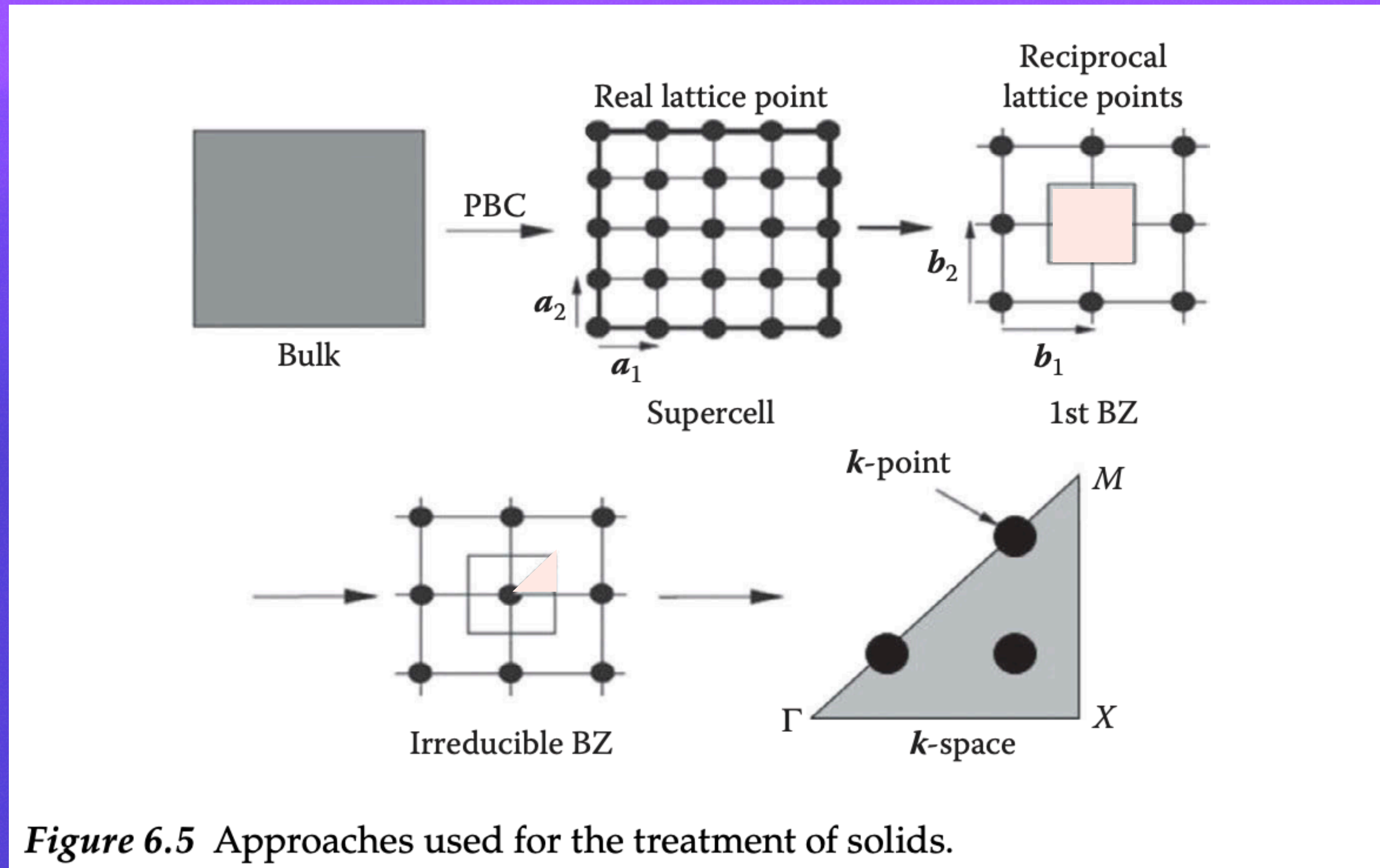
Symmetry and Periodicity

Reciprocal Space- Brillouin Zone



Symmetry and Periodicity

Reciprocal Space- Irreducible Brillouin Zone



Symmetry and Periodicity

Reciprocal Space- Brillouin Zone

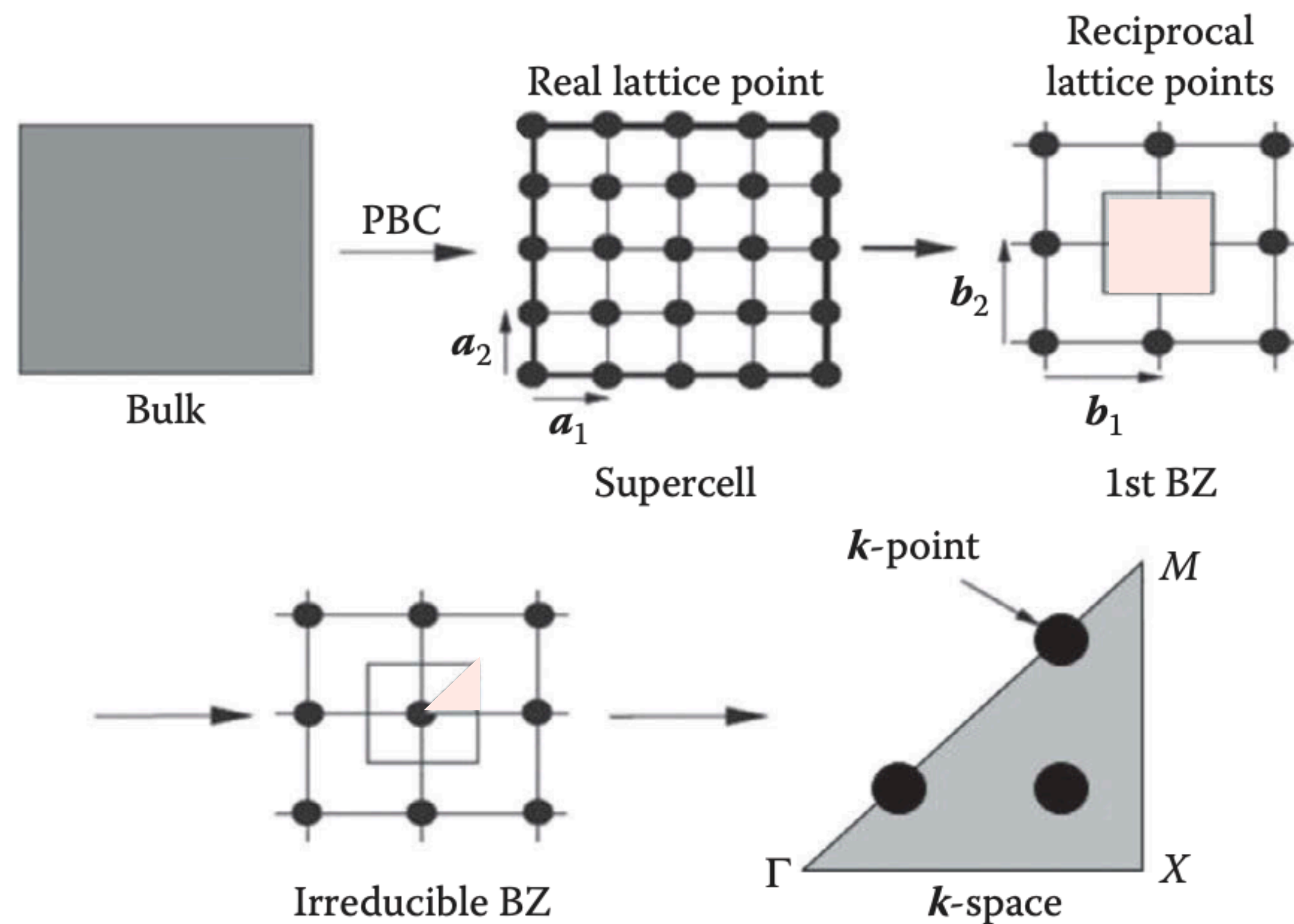


Figure 6.5 Approaches used for the treatment of solids.

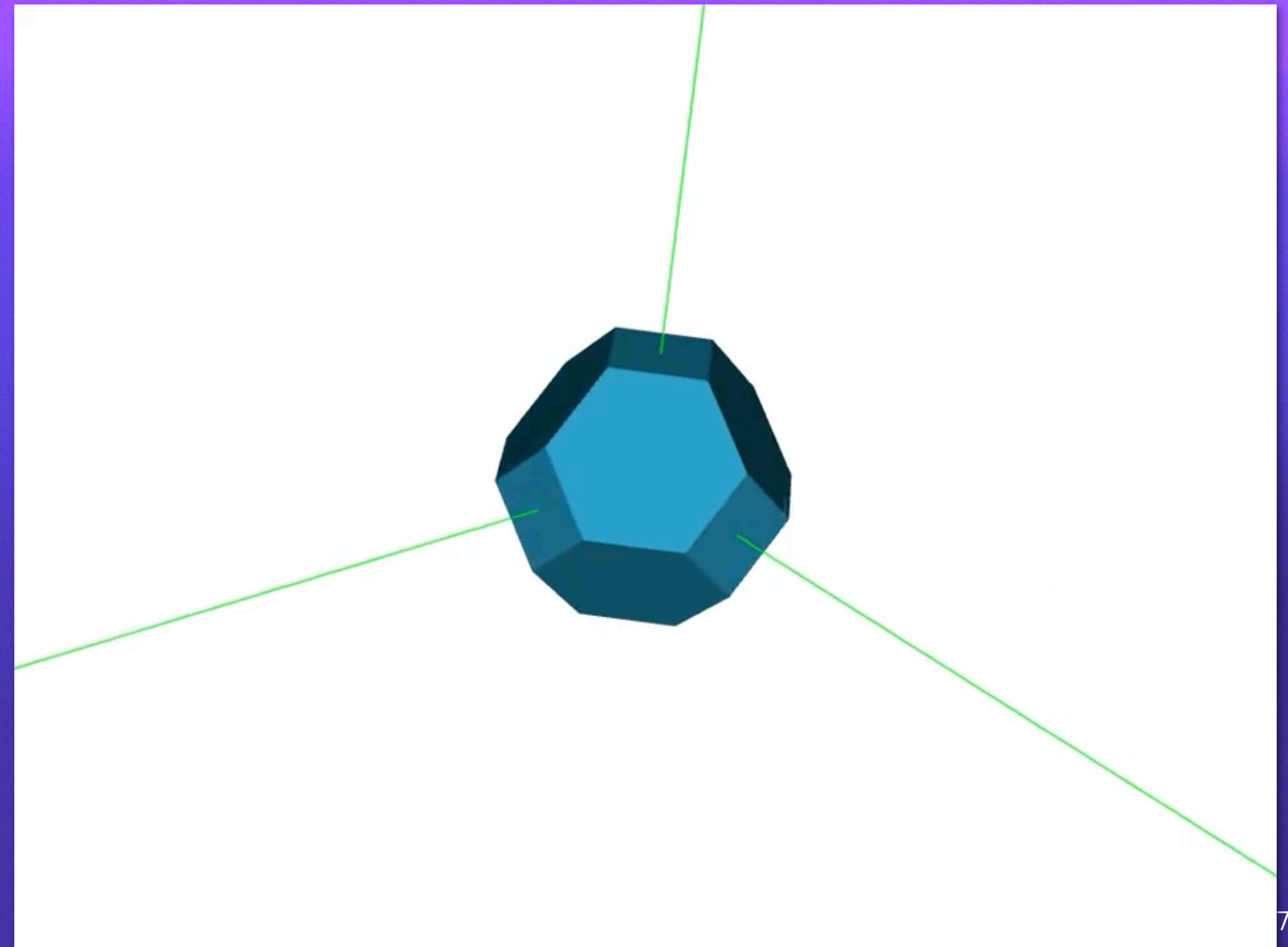
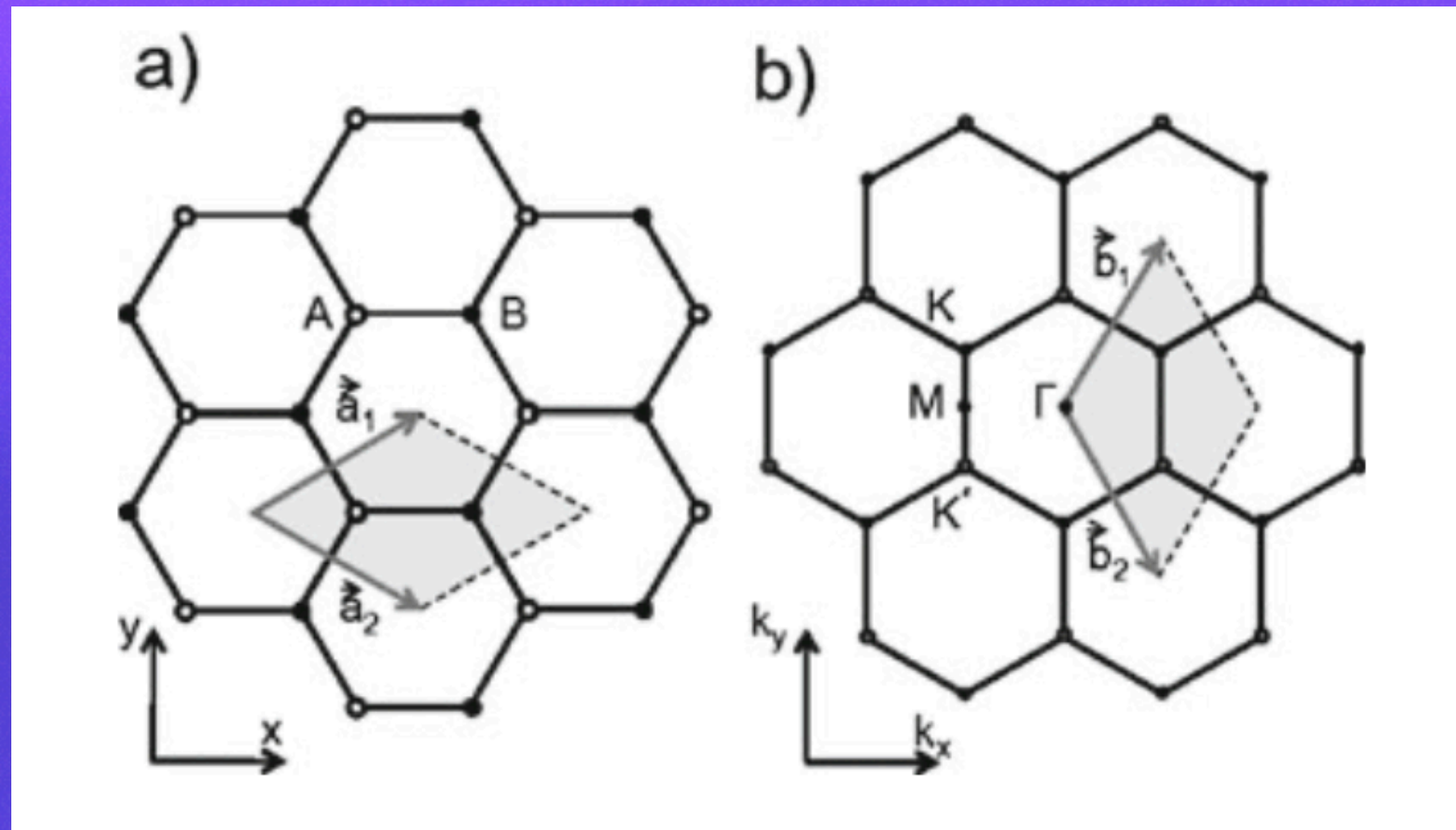
The 1st BZ is reduced to IBZ by symmetry operations without losing any information.

- The IBZ is mapped with discrete k-points, and all necessary quantities are obtained by integration/summation/extrapolation on these points.



Symmetry and Periodicity

Reciprocal Space for different symmetries



Bloch Theorem

Bloch Theorem

A consequence of the periodicity

- Periodicity in the real lattice ✓
- Periodicity in the potential ✓
- Periodicity of the density ✓
- Periodicity of the wave function? ✗



I Can do it!

Valence Electrons like 'free electrons'

$$\Psi(\mathbf{r}) = C \exp(i\mathbf{k} \cdot \mathbf{r})$$

Solution 'modulated' by a periodicity

$$u_k(\mathbf{r}) = u_k(\mathbf{r} + \mathbf{R}), \text{ then}$$

$$\Psi_k(\mathbf{r}) = u_k(\mathbf{r}) \exp(i\mathbf{k} \cdot \mathbf{r})$$

$$\Psi_k(\mathbf{r} + \mathbf{R}) = \Psi_k(\mathbf{r}) \exp(i\mathbf{k} \cdot \mathbf{R})$$

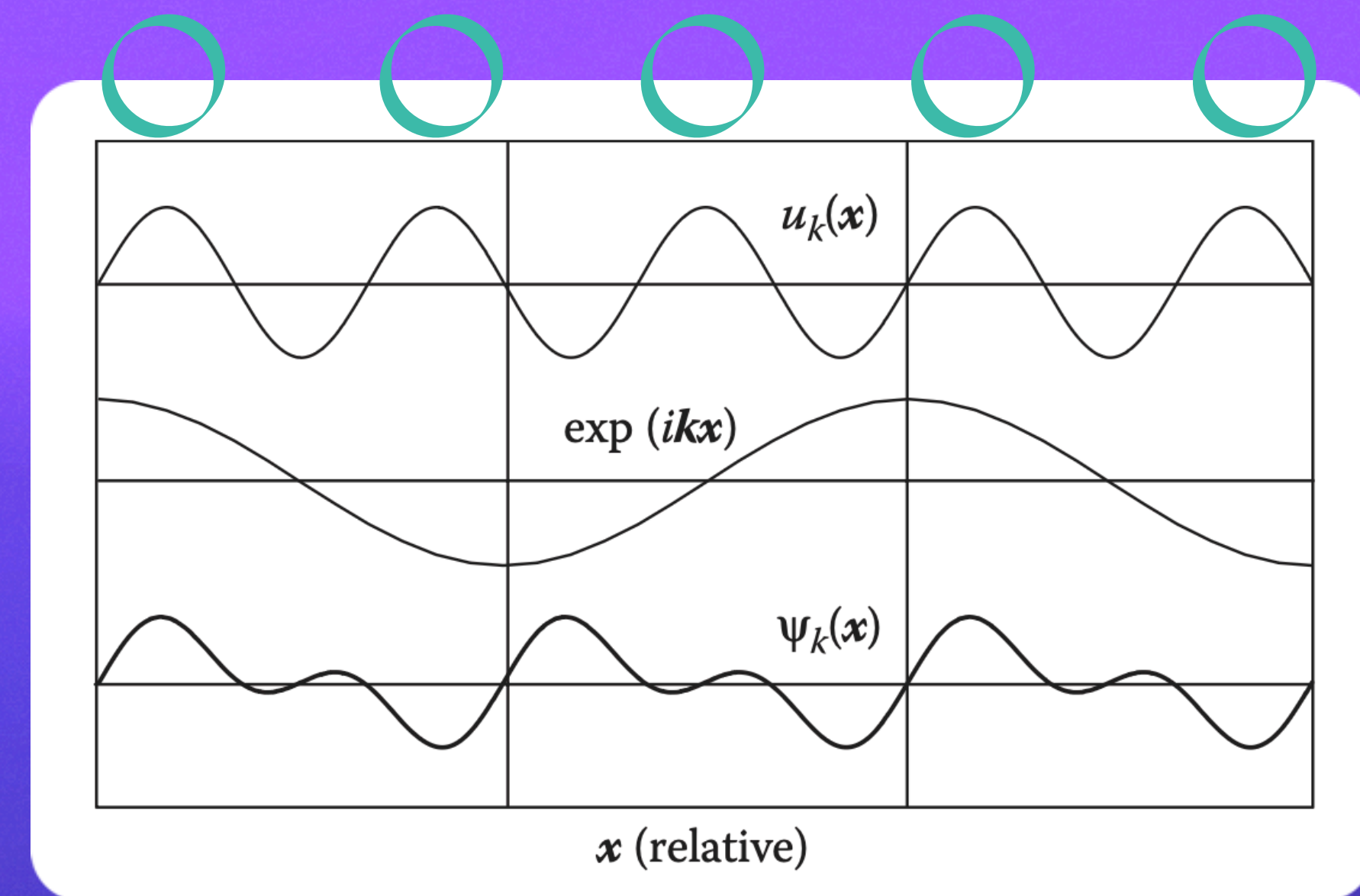
Bloch Theorem

A consequence of the periodicity

- Periodicity in the real lattice ✓
- Periodicity in the potential ✓
- Periodicity of the density ✓
- Periodicity of the wave function? ✗



I Can do it!



$$\Psi_k(\mathbf{r}) = u_k(\mathbf{r})\exp(i\mathbf{k} \cdot \mathbf{r})$$

$$\Psi_k(\mathbf{r} + \mathbf{R}) = \Psi_k(\mathbf{r})\exp(i\mathbf{k} \cdot \mathbf{R})$$

Bloch Theorem

Expansion Fourier

$u_k(\mathbf{r})$ is a periodic function...

$$u_k(\mathbf{r}) = \sum_G c_k(G) \exp(i\mathbf{G} \cdot \mathbf{r})$$

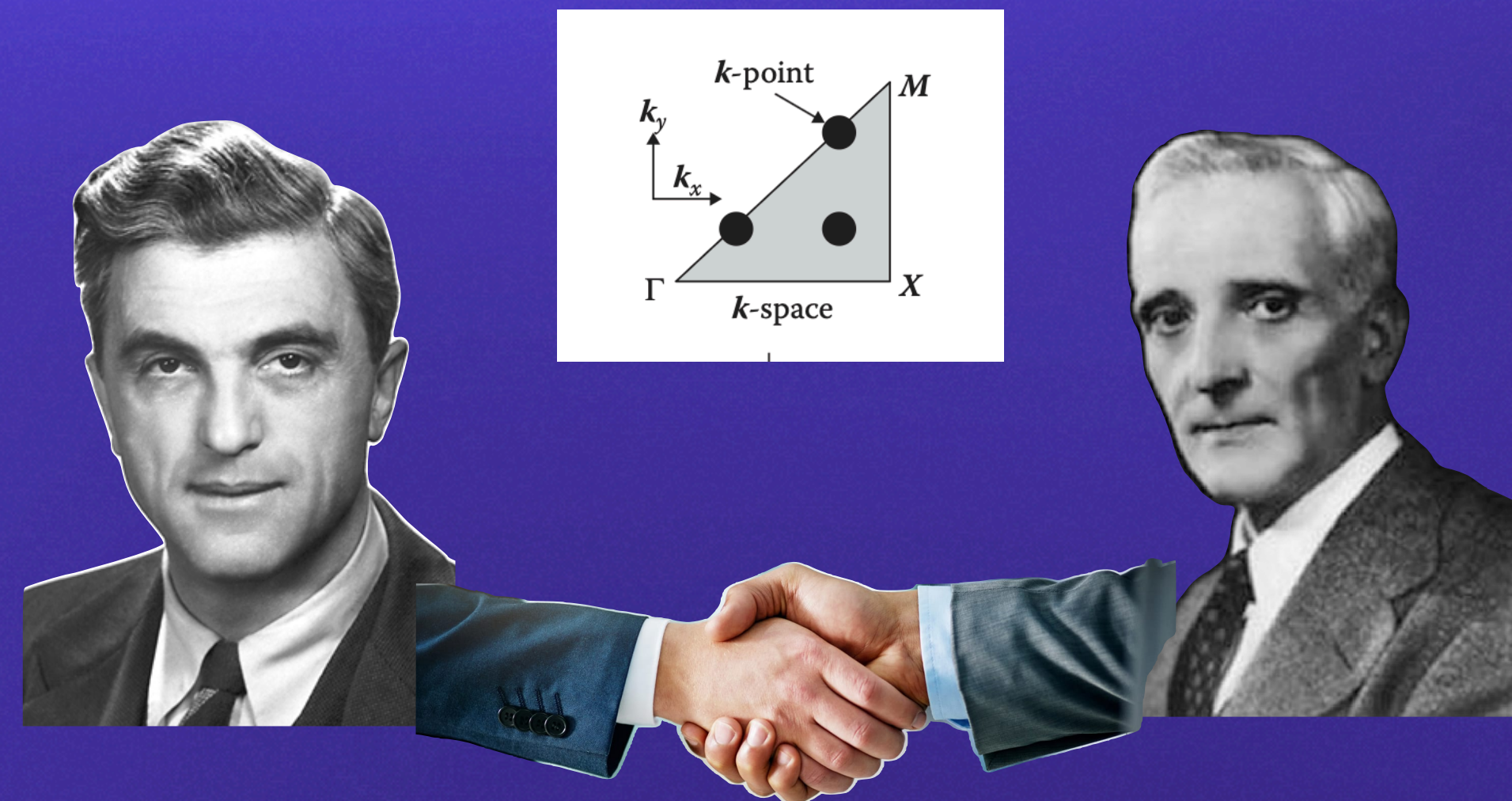
Then...

$$\Psi_k(\mathbf{r}) = u_k(\mathbf{r}) \exp(i\mathbf{K} \cdot \mathbf{r})$$

$$\Psi_k(\mathbf{r}) = \sum_G c_k(G) \exp[i(\mathbf{G} + \mathbf{K}) \cdot \mathbf{r}]$$

This ensures the expansion perfectly matches the crystal's periodicity, because $\mathbf{G}$ are the reciprocal lattice vectors.

The final result is that any electron state in a crystal can be expressed as a simple sum of plane waves. This makes the problem computationally feasible and is the foundation for most electronic structure calculations, like DFT.



Bloch Theorem

KS orbitals

Same Bloch idea, a periodic contribution

$$\phi_{jk}(\mathbf{r}) = \sum_{\mathbf{R}} \exp(i\mathbf{k} \cdot \mathbf{r}) \phi_j(\mathbf{r} - \mathbf{R})$$

Now we can transform the Schrödinger equation to reciprocal space:

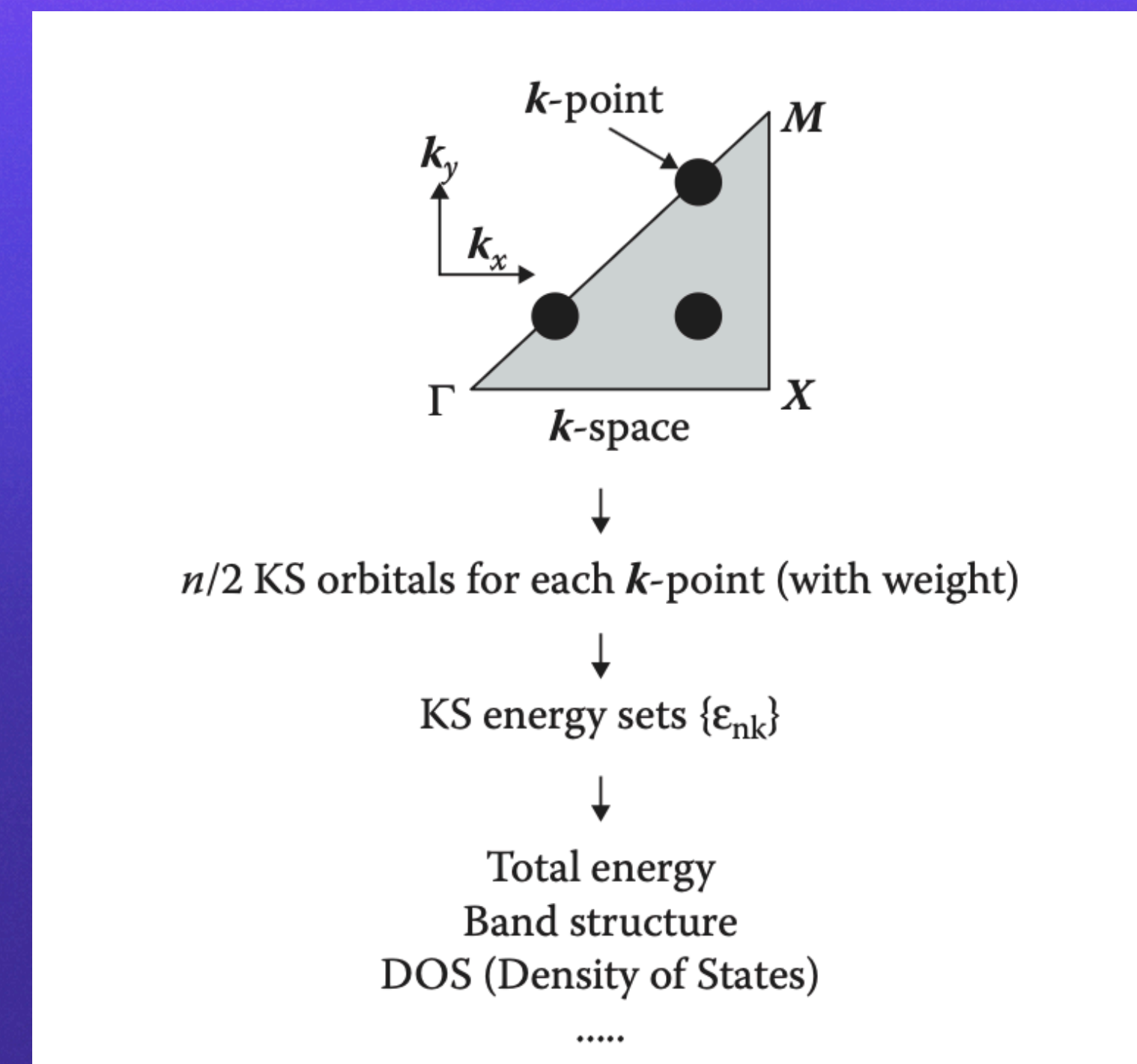
$$\hat{H}\Psi_{\alpha}(r) = E_{\alpha}\Psi_{\alpha}(r)$$

$$\sum_j c_{j\alpha} \hat{H} \phi_{jk}(r) = E_{\alpha} \sum_j c_{j\alpha} \phi_{jk}(r)$$

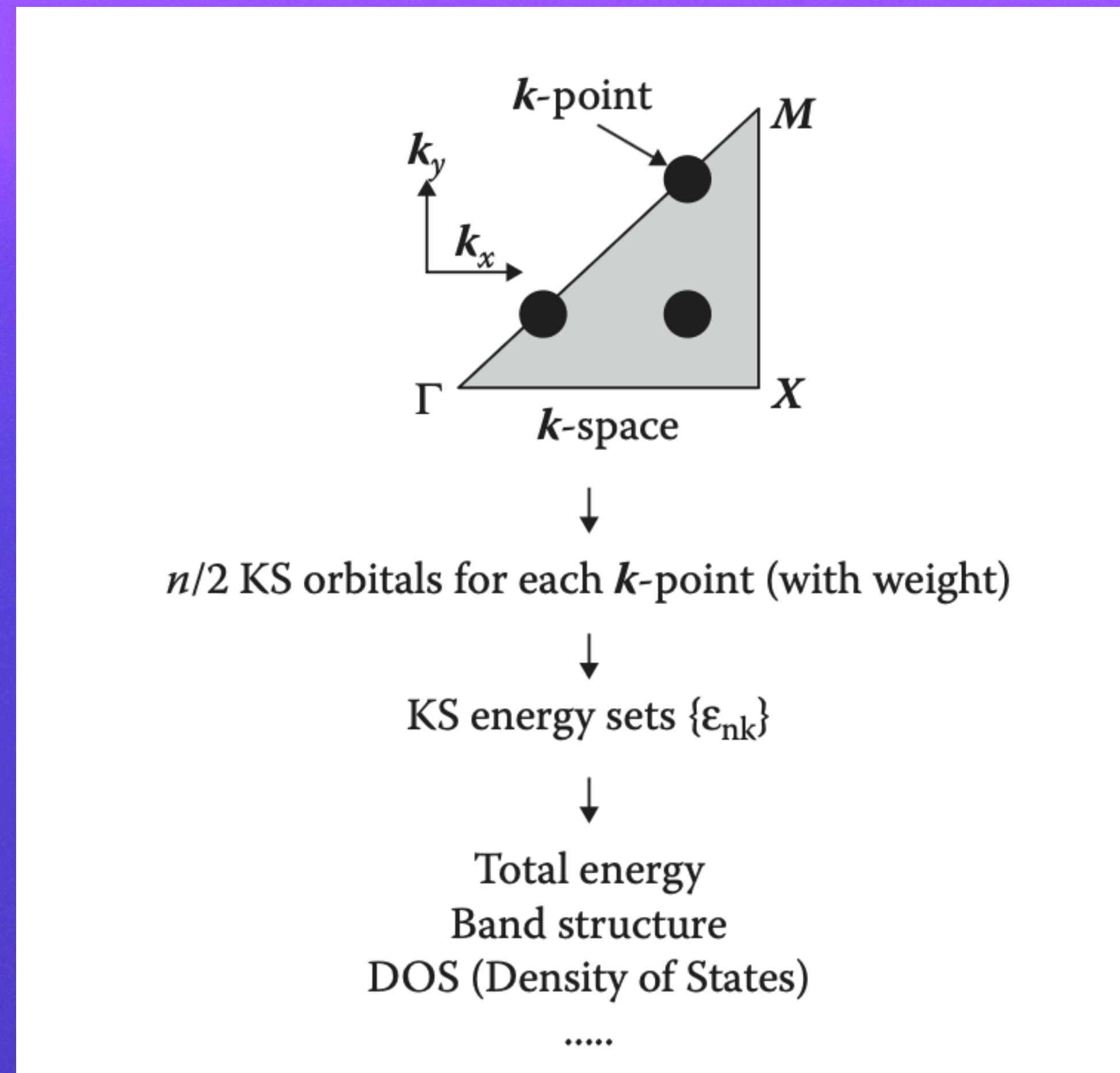
$$\sum_j c_{j\alpha} H_{lj}(\mathbf{k}) = E_{\alpha} \sum_j c_{j\alpha} S_{lj}(\mathbf{k})$$

The problem is decoupled. We solve a small, independent equation for each **k**-point of interest in the Brillouin Zone.

$$|H(k_i) - E_{\alpha}S(k_i)| = 0$$

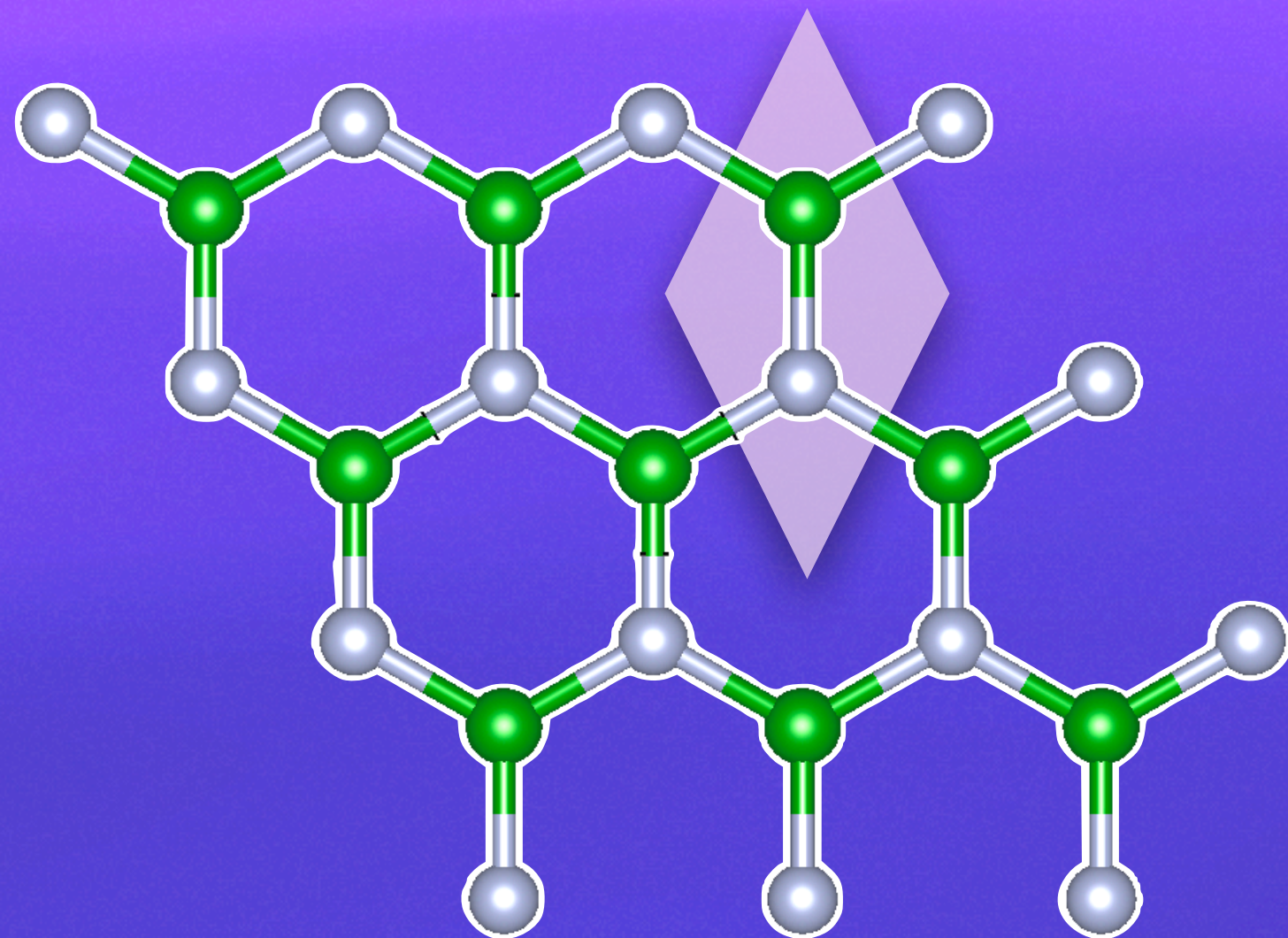


Band Structure



Band Structure

hBN Case



Boron (B): 3 valence electrons

Nitrogen (N): 5 valence electrons

Unit Cell: Contains 1 Boron atom and 1 Nitrogen atom.

$3 + 5 = 8$ valence electrons per unit cell.

8 bands!

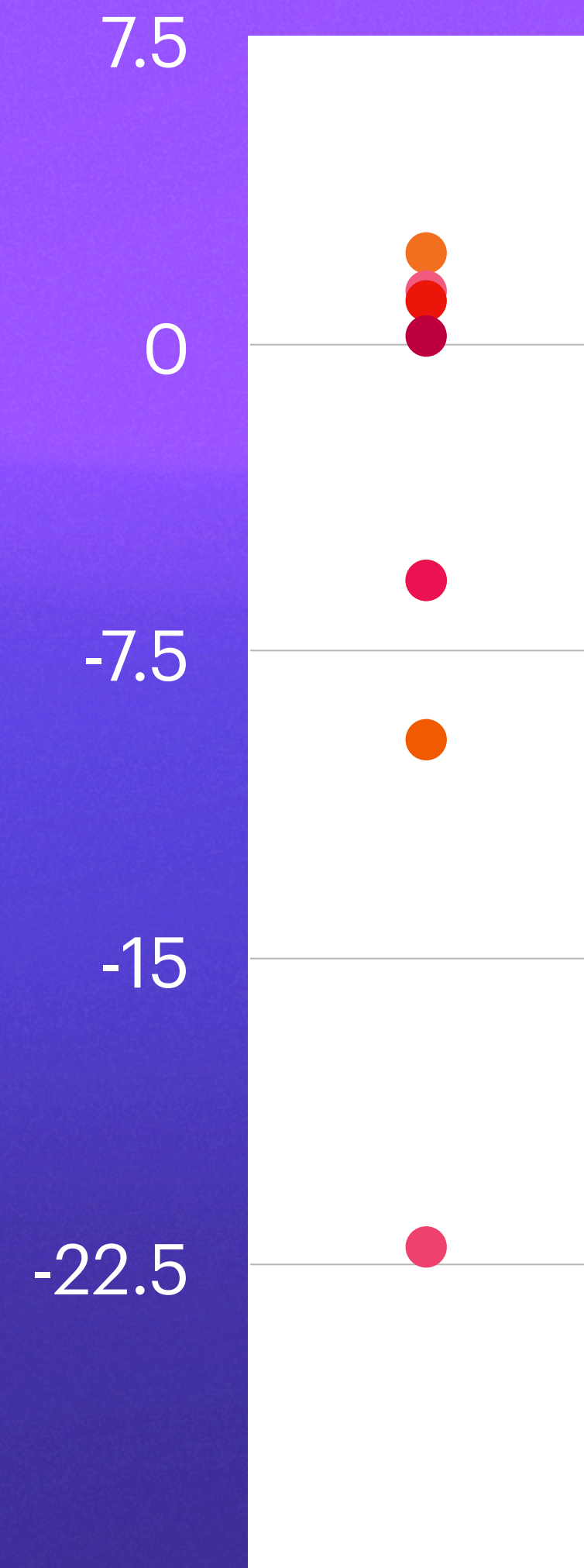
$N/2 \text{ KS} = 4$ occupied bands

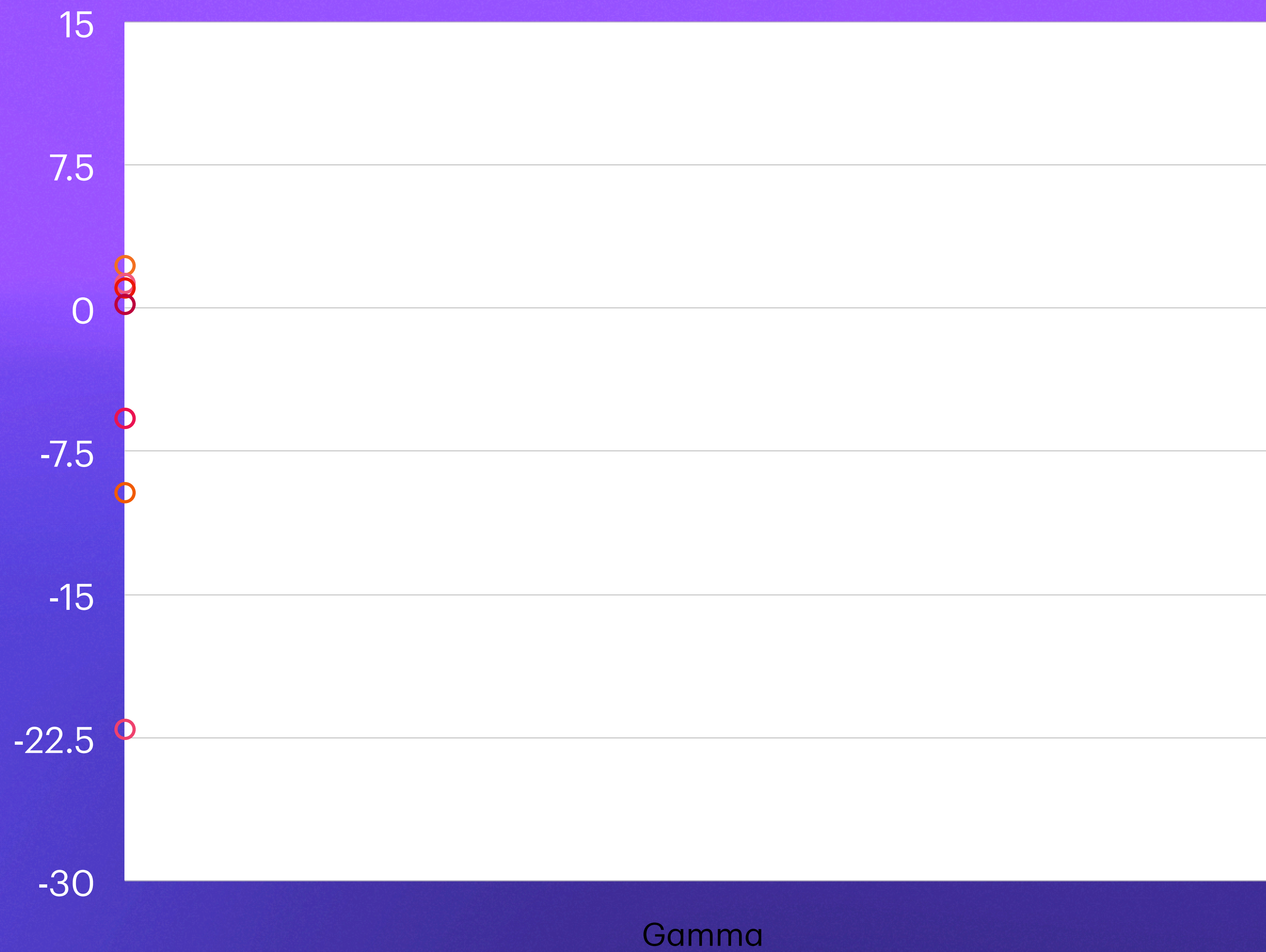
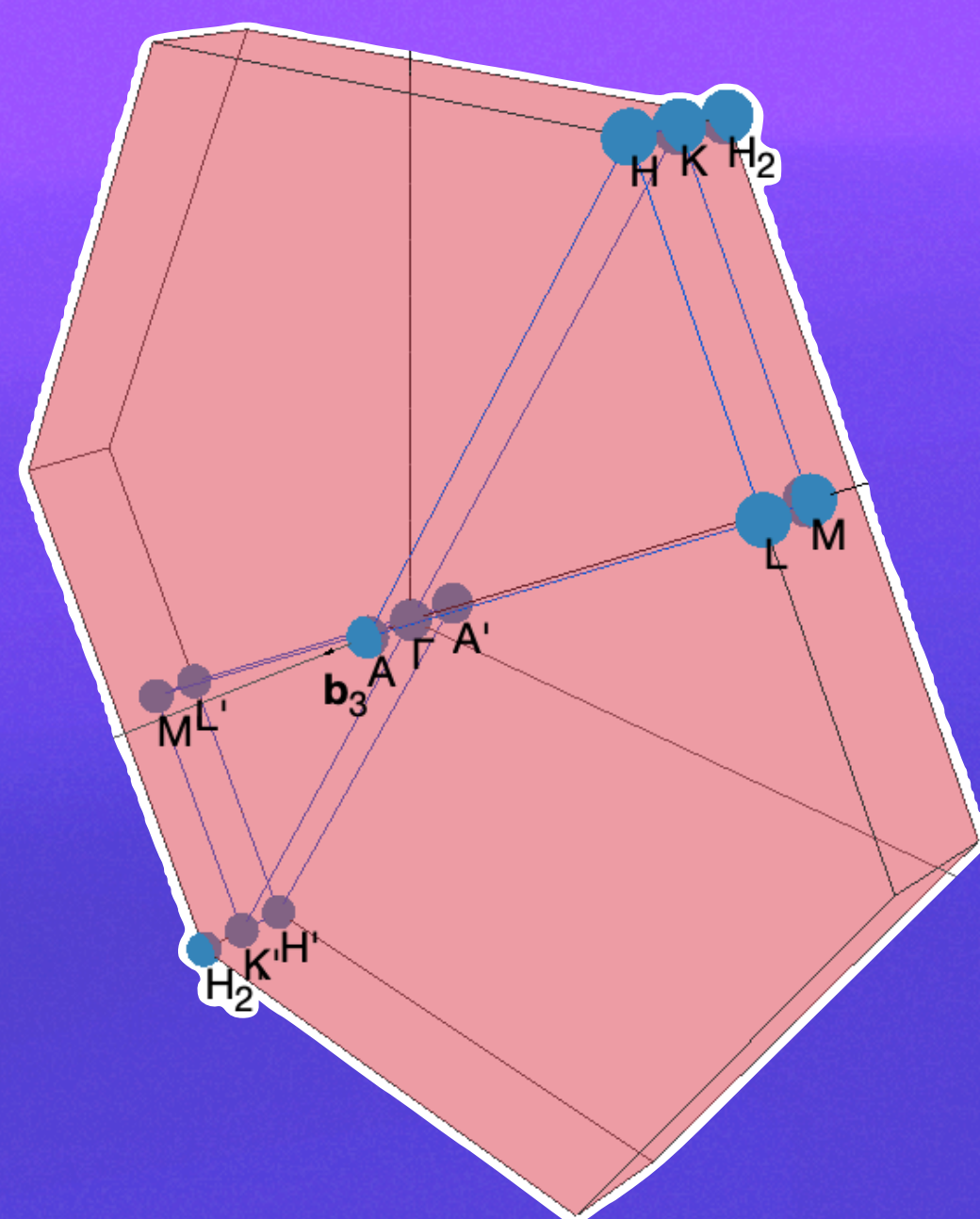
Band Structure

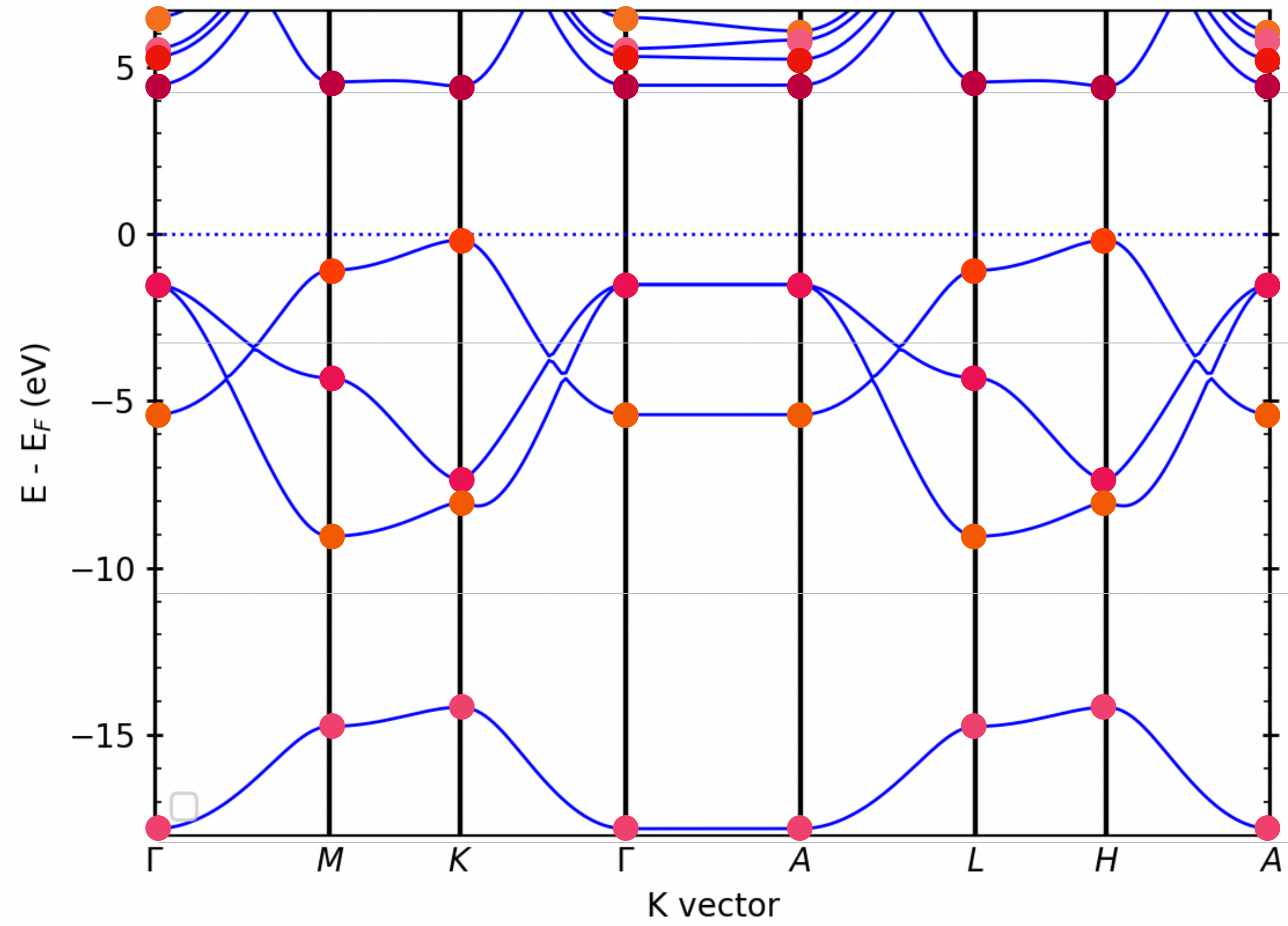
Energy calculation for Γ point

The problem is decoupled. We solve a small, independent equation for each $\mathbf{k}$ -point of interest in the Brillouin Zone.

$$|H(k_i) - E_\alpha S(k_i)| = 0$$

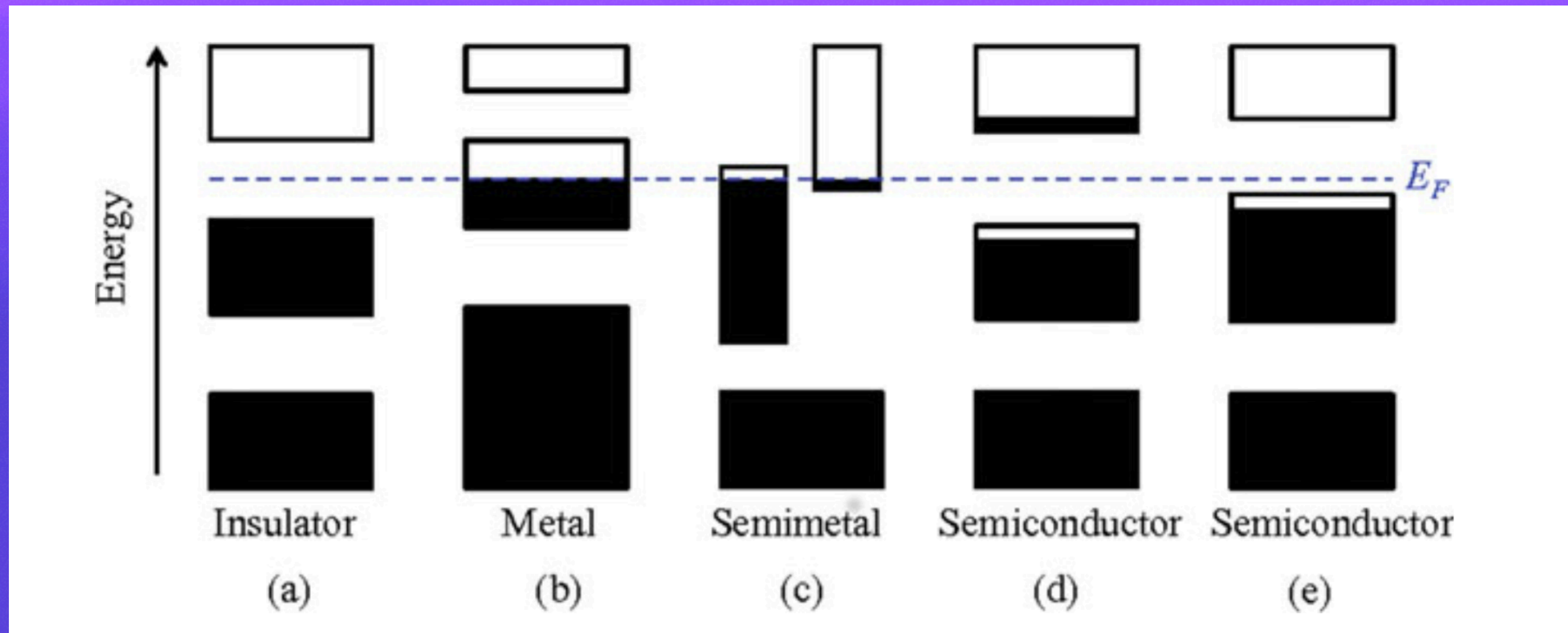






Band Structure

Interpretación



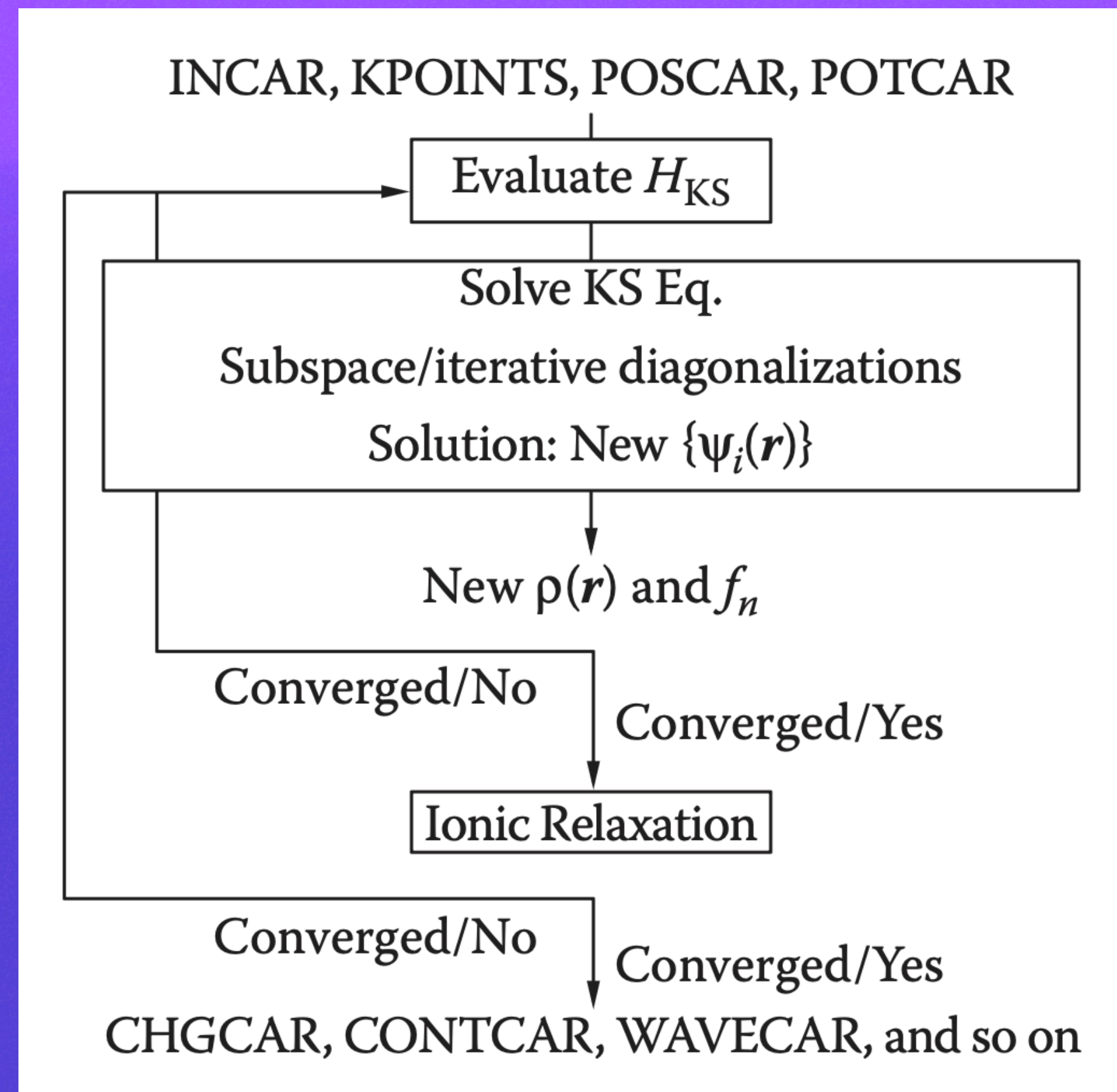
Coffee Break

Hands-on: VASP

Hand-on: VASP

Inputs files

- POSCAR
- INCAR
- KPOINTS
- POTCAR



Step 0: Perform convergence tests to correctly prepare the inputs.

Step 1: Ionic Relaxation

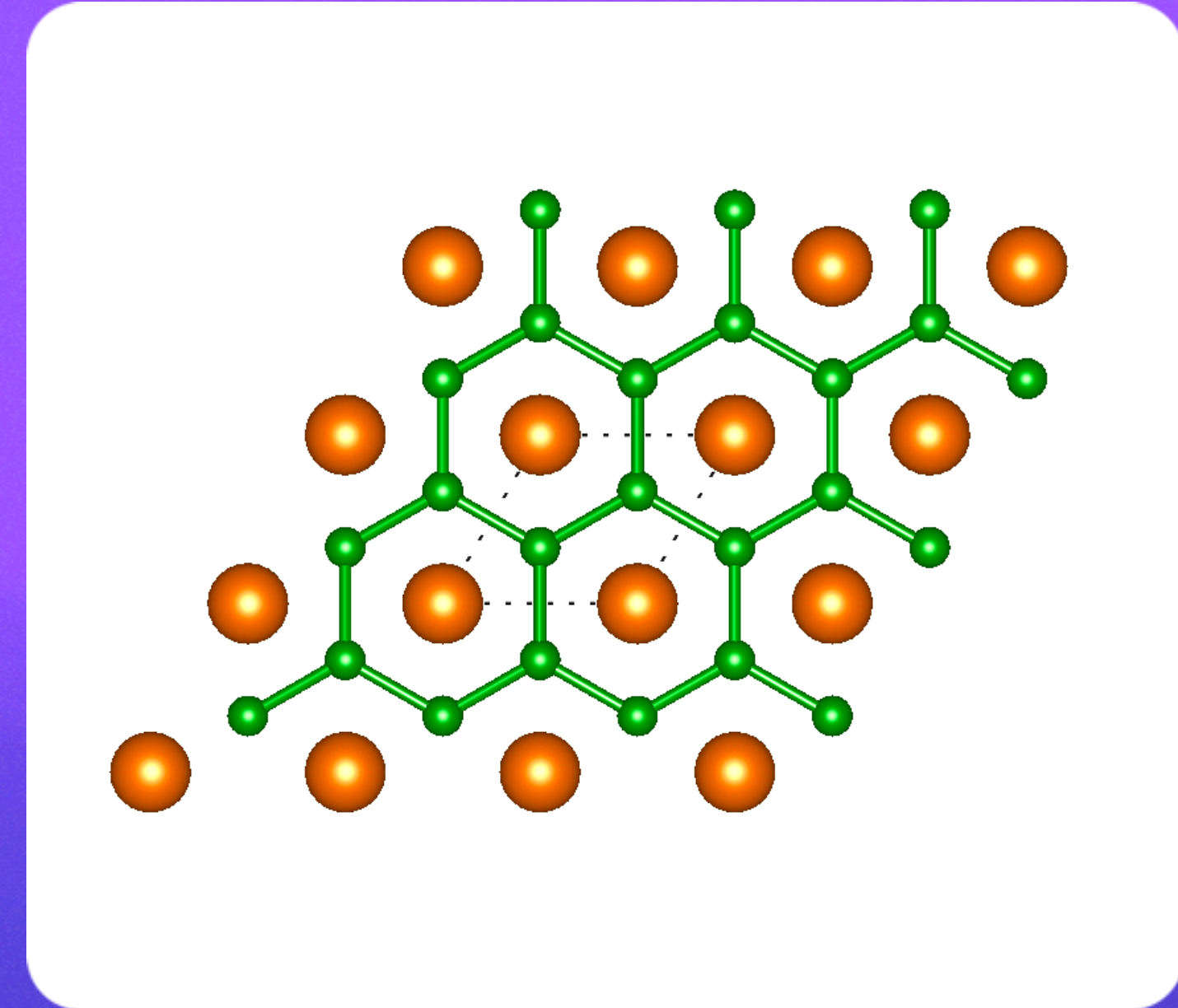
Step 2: SCF for the CHGCAR

Step 3: NSCF for bands

Hand-on: VASP

MgB₂

- Hexagonal Lattice
- Spacial group P6/mmm No. 191
- 3 atoms for unit cell
- Superconductor



The Materials Project

NOMAD

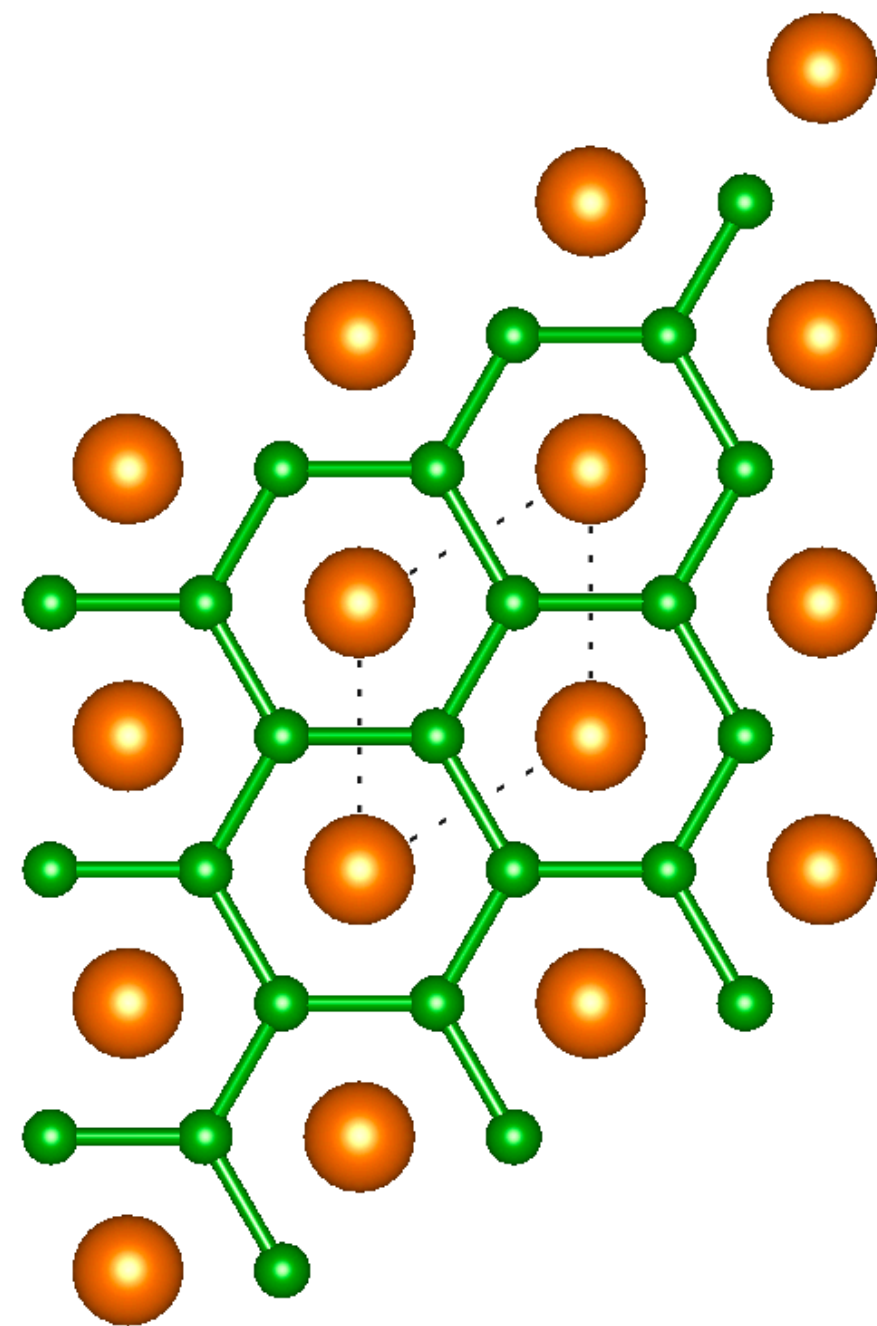
OQMD
The Open Quantum
Materials Database

NEMAD

North East Materials Database

Hand-on: VASP

POSCAR



```
[student25@leftraru2 scf]$ less POSCAR
```

Mg1B2

1.0000000000000000

1.5357094241588196 -2.6599267483054190 0.0000000000000000

1.5357094241588196 2.6599267483054190 0.0000000000000000

0.0000000000000000 0.0000000000000000 3.5308220899090061

Mg B

1 2

Direct

-0.0000000000000000 -0.0000000000000000 0.0000000000000000

0.6666666666666643 0.3333333333333357 0.5000000000000000

0.3333333333333357 0.6666666666666643 0.5000000000000000

Hand-on: VASP

POTCAR

Standard (Recommended)

- Includes the most important valence electrons.
- Suitable for most general-purpose calculations.
- Offers a good balance between accuracy and computational cost.

"Hard" & Specialized (**_sv**, **_pv**, **_h**, **_GW**)

- Provide higher accuracy by including more electrons (semicore) in the valence shell.
- They are significantly more computationally expensive.

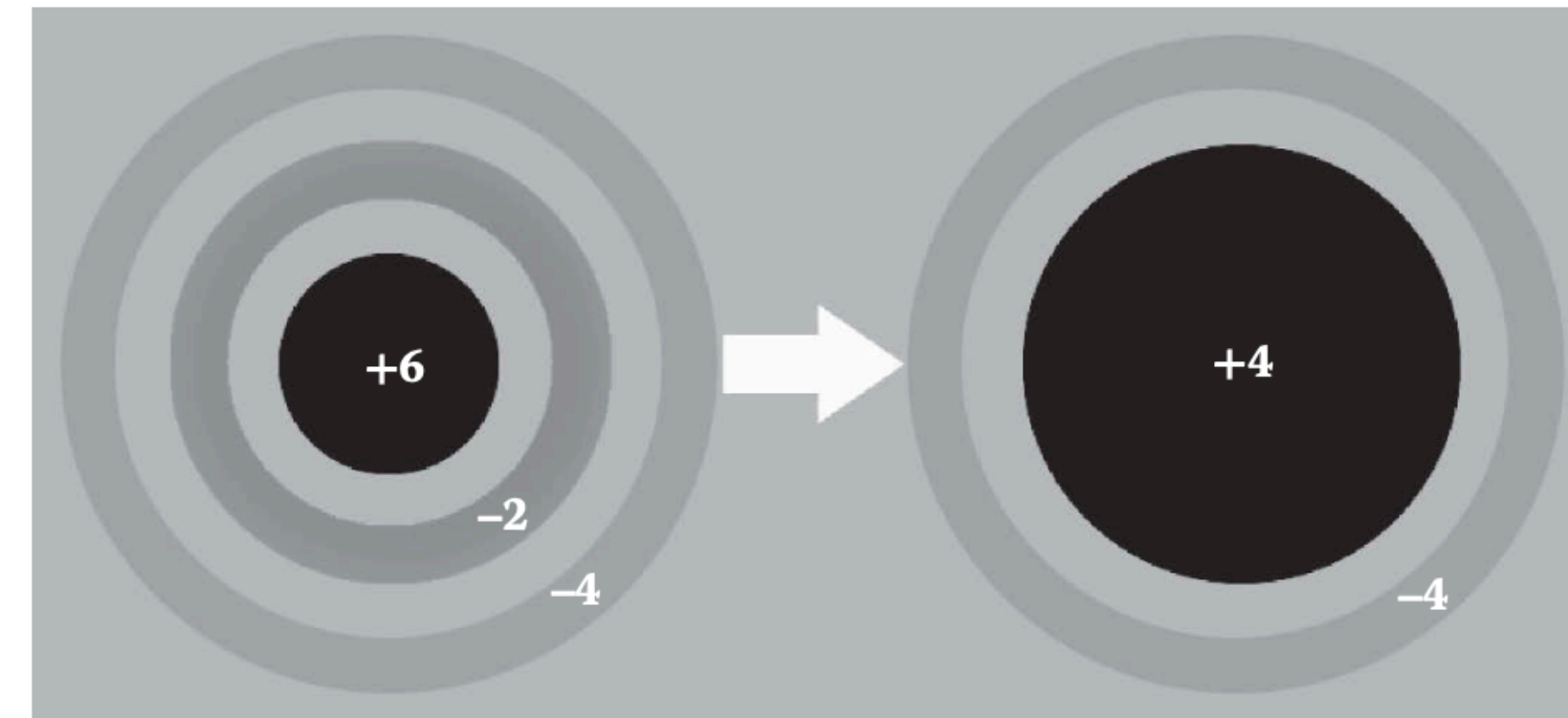


Figure 6.2 Atomic system of carbon showing the frozen core and valence electrons for the construction of a pseudopotential.

```
PAW_PBE Mg 13Apr2007
2.0000000000000000
parameters from PSCTR are:
SHA256 = a90315392b5cf46bd986ff2010bfd53a40dec98e36f8d1f6277412e9d0e81a9f Mg/POTCAR
COPYR = (c) Copyright 13Apr2007 Georg Kresse
COPYR = This file is part of the software VASP. Any use, copying, and all other rights are regulated by
the VASP license agreement.
COPYR = If you do not have a valid VASP license, you may not use, copy or distribute this file.
VRHFIN =Mg: s2p0
LEXCH = PE
EATOM = 23.0369 eV, 1.6932 Ry
...
POMASS = 24.305; ZVAL = 2.000 mass and valenz
RCORE = 2.000 outmost cutoff radius
RWIGS = 2.880; RWIGS = 1.524 wigner-seitz radius (au A)
ENMAX = 200.000; ENMIN = 100.000 eV
```

Convergence Tests

Hand-on: VASP

ENCUT problem - Where do I stop summing?

$$\psi_k(r) = \sum_{|k+G| < cut} c_{k+G} \exp[i(k+G) \cdot r]$$

We only include in our sum those plane waves whose kinetic energy is less than E_{CUT} .

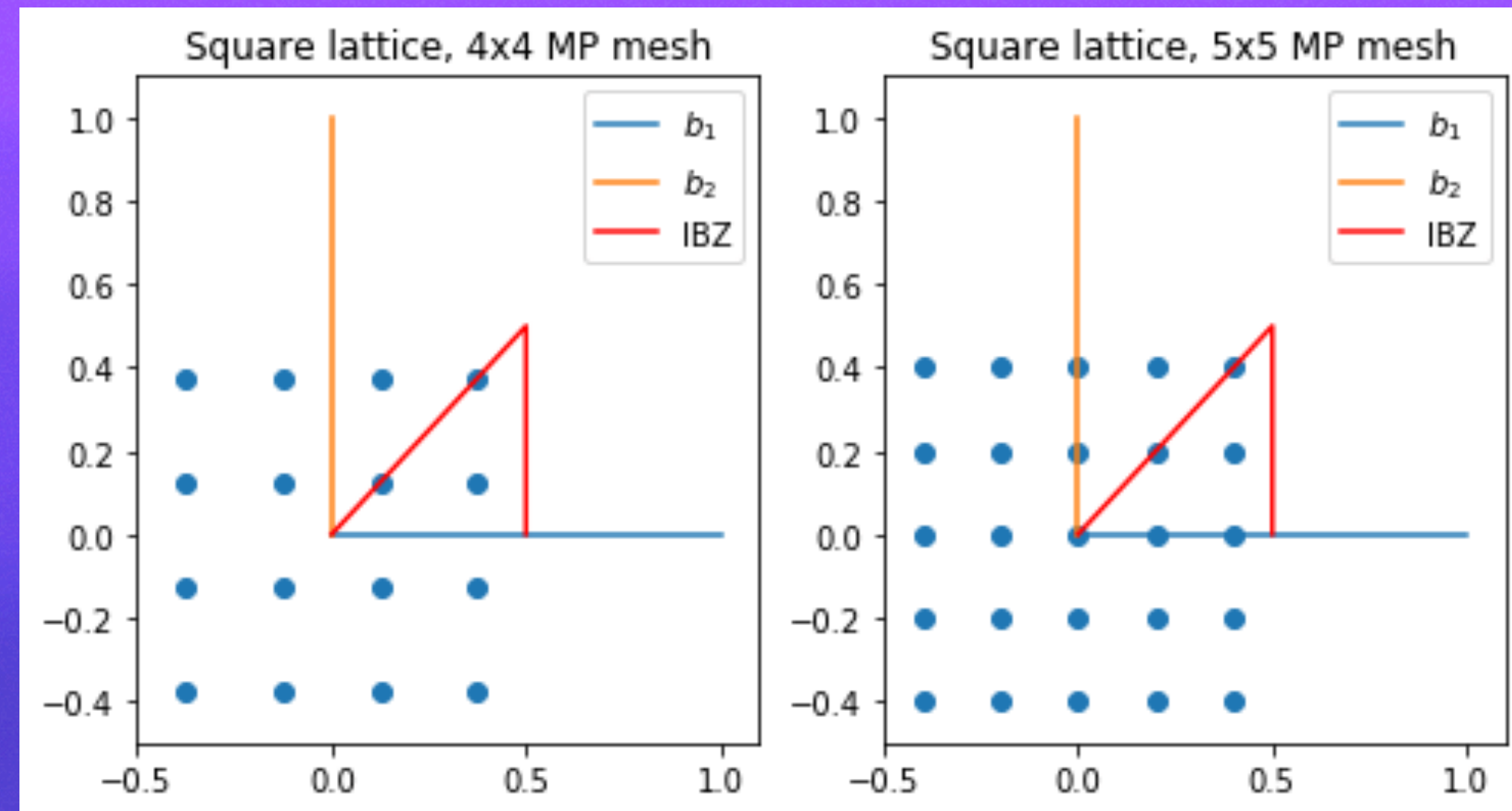
$$E_{kin} = \frac{1}{2} |k+G|^2 < E_{CUT}$$

```
ISTART = 0      ! doesn't read the WAVECAR file
ENCUT = 500     ! energy cutoff for the plane-wave basis in eV
ISMear = 0      ! method for the partial occupancies on each
                 ! orbital (=0 Gaussian smearing)
SIGMA = 0.01    ! width of the smearing in eV
EDIFF = 1e-8    ! global break condition for the electronic SC-
                 ! loop in eV

LORBIT = 11
LREAL = False
LWAVE = False
NPAR = 2
```

Hand-on: VASP

KPOINTS - How to star?



Regular 4 x 4 x 3 mesh centered #linea de comentario

0

Gamma

4 4 3

Given reciprocal vector lengths with a ratio of $0.4 : 0.4 : 0.2$:

- The normalized ratio is calculated by dividing by the smallest value (0.2), which gives $2 : 2 : 1$.
- **Rule:** We need more k-points for longer reciprocal vectors to maintain a uniform sampling density.
- **Conclusion:** The k-point mesh should follow this $2 : 2 : 1$ ratio (e.g., $8 \times 8 \times 4$, $10 \times 10 \times 5$, etc.).

Hand-on: VASP

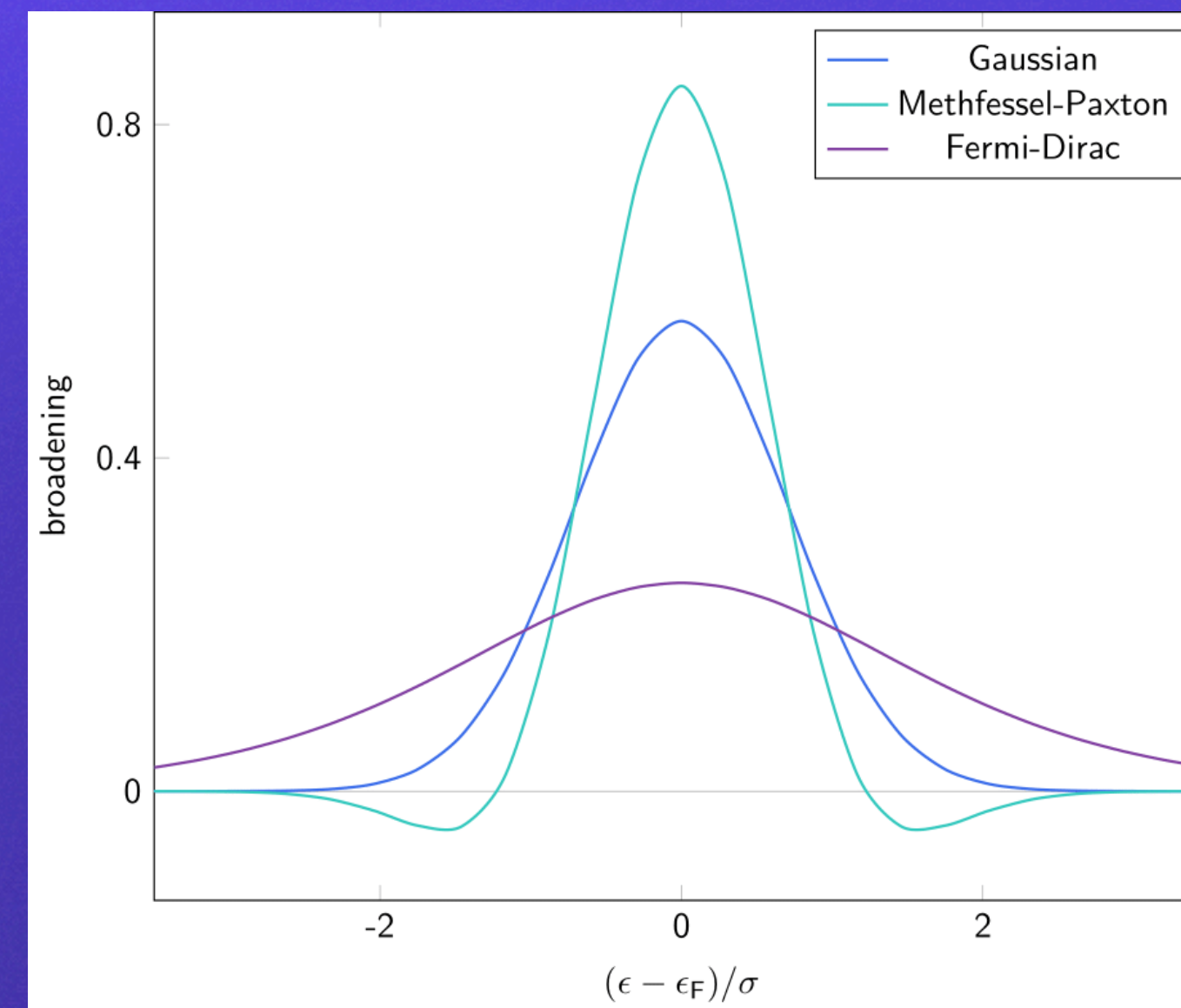
Other important tag: Smearing

Problem in Metals: Electron occupation drops abruptly ($1 \rightarrow 0$) at the Fermi level, requiring dense and computationally expensive k-point meshes to converge.

Solution (Smearing): It smears the occupation of electrons near the Fermi level to improve the stability and speed of the calculation.

```
ISTART = 0      ! doesn't read the WAVECAR file
ENCUT = 500     ! energy cutoff for the plane-wave basis in eV
ISMEAR = 0      ! Util si no sabes que es, 1 y 2 for metals:
method of Methfessel-Paxton
SIGMA = 0.01    ! width of the smearing in eV
EDIFF = 1e-8    ! global break condition for the electronic SC-
loop in eV

LORBIT = 11
LREAL = False
LWAVE = False
NPAR = 2
```



Smearing technique

Page [Discussion](#)

How to set ISMEAR

Hand-on: VASP

Structural Relaxation

Prerequisites: Ensure **ENCUT** and **KPOINTS** are converged from previous static calculations.

INCAR Setup: Add new tags required for relaxation (e.g., **IBRION**, **NSW**, **ISIF**).

Post-Relaxation Checklist: Review **OUTCAR** for convergence, **CONTCAR** for the new atomic positions, and **CHGCAR** for the final charge density.

```
SYSTEM = RELAX (ENCUT = 300, 12x12x9)
ISTART = 0          ! don't read WAVECAR
ENCUT = 300        ! energy cutoff
ISMEAR = 0         ! Gaussian smearing
SIGMA = 0.2        ! width of the smearing
EDIFF = 1e-6       ! electronic convergence

LORBIT = 11
LREAL = False
LWAVE = False

# Relax
NSW = 10           ! max number of ionic steps
IBRION = 2         ! ionic relaxation (conjugate gradient algorithm)
ISIF = 3           ! relajar todo
EDIFFG = -1E-2     ! break condition for the ionic relaxation loop
NELM = 100         ! max number of electronic SC steps
```

Hand-on: VASP

Bands Calculation

Prerequisites: Start with a converged self-consistent calculation (SCF or relaxation). You must have the converged **CHGCAR** file.

INCAR Setup: Modify the **INCAR** for a non-self-consistent run. Key tags include **ICHARG = 11** (to read the fixed charge density) and an appropriate **ISMEAR** for band structures (e.g., **ISMEAR = 0**).

```
SYSTEM = RELAX (ENCUT = 300, 12x12x9)
ISTART = 0          ! don't read WAVECAR
ENCUT = 300 ! energy cutoff
ISMEAR = 0          ! Gaussian smearing
SIGMA = 0.1         ! width of the smearing
EDIFF = 1e-6        ! electronic convergence

LORBIT = 11
LREAL = False
LWAVE = False

# Relax
#NSW = 10           ! max number of ionic steps
#IBRION = 2         ! ionic relaxation (conjugate gradient algorithm)
#ISIF = 3           ! relajar todo
#EDIFFG = -1E-2     ! break condition for the ionic relaxation loop
#NELM = 100         ! max number of electronic SC steps

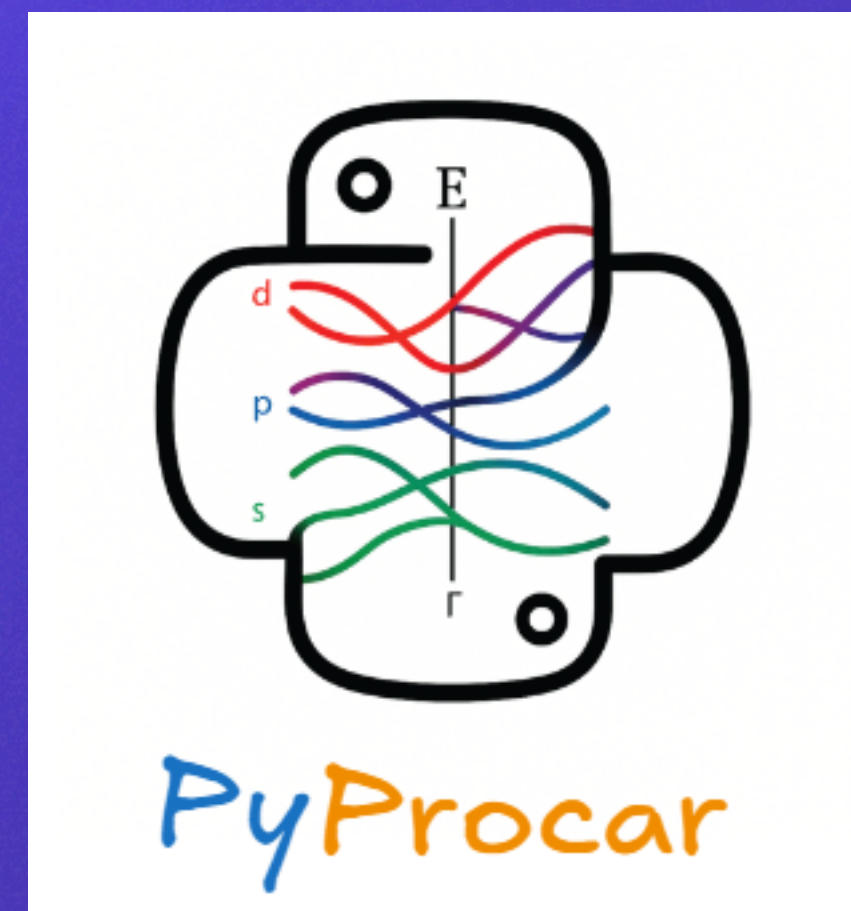
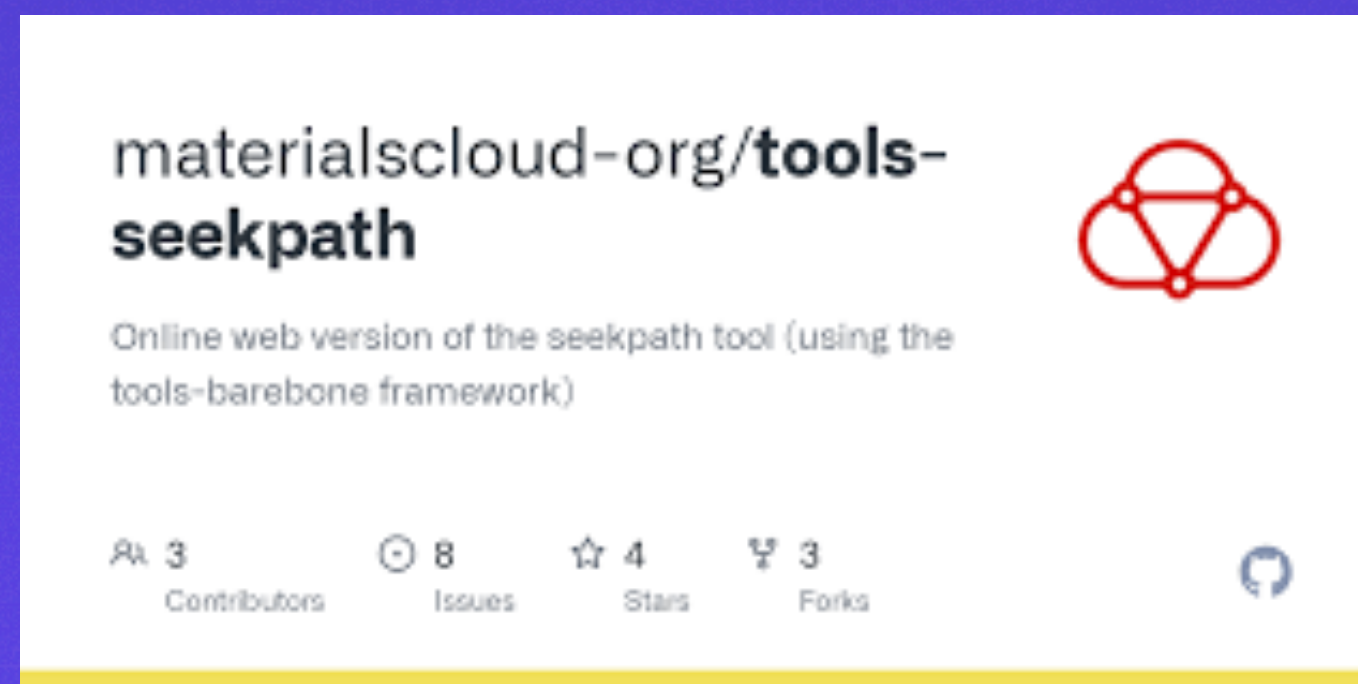
ICHARG=11
```

Hand-on: VASP

Bands Calculation

KPOINTS File: Replace the standard k-point mesh with a new file defining the path along high-symmetry lines in the Brillouin Zone.

Post-Calculation Checklist: Use Pyprocar



```
K-Path Gamma-M-L-A-T
40
Line-mode
reciprocal
0.0000000000 0.0000000000 0.0000000000 Gamma
0.5000000000 0.0000000000 0.0000000000 M

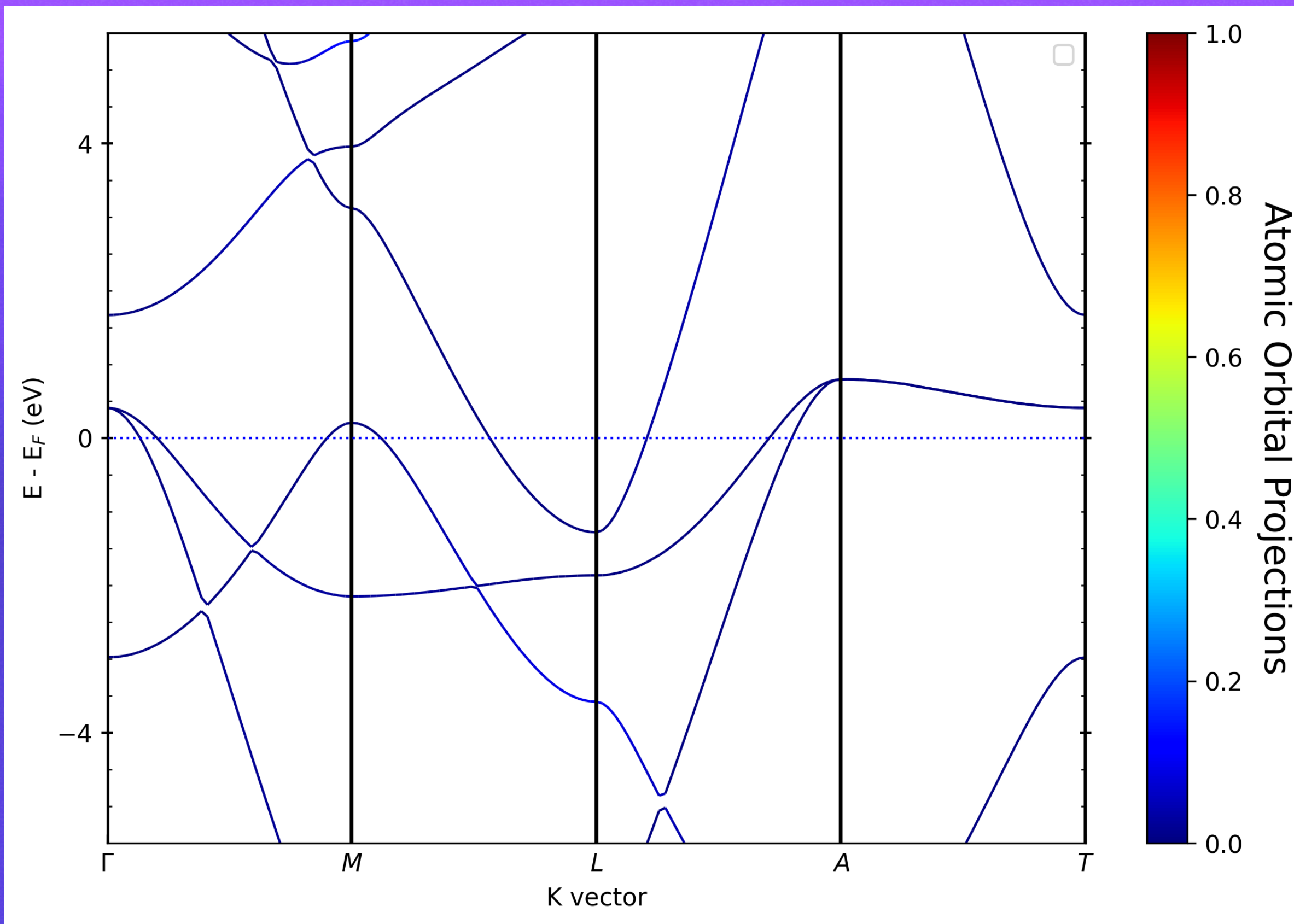
0.500      0      0      M
0.5        0      0.5    L

0.5000000000 0.0000000000 0.5000000000 L
0.0000000000 0.0000000000 0.5000000000 A

0      0      0.5      A
0.0000000000 0.0000000000 0.0000000000 T
```

Hand-on: VASP

Bands Calculation- Post processing



```
import pyprocar
import numpy as np
import matplotlib.pyplot as plt

fermi_energy= 5.3669
pyprocar.bandsplot(code='vasp',
                    dirname='./',
                    elimit=[-8.5,8.5],
                    mode='plain',
                    color='pink',
                    fermi= fermi_energy,
                    savefig='./bands1.png',
                    show=False,
                    print_plot_opts=True)

pyprocar.bandsplot(code='vasp',
                    dirname='./',
                    elimit=[-5.5,5.5],
                    mode='parametric',
                    atoms=[0],
                    orbitals=[1,3],
                    fermi= fermi_energy,
                    savefig='./bands2.png',
                    show=False,
                    print_plot_opts=True)
```